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1 Precision Measurement of the Thrust Distribution and α_s Extraction in e^+e^- Collisions at $\sqrt{s} = 91.2$ GeV Using Archived ALEPH Data

Analysis: thrust_alphas **Collaboration:** ALEPH (archived data re-analysis) **Phase:** 5 — Final Analysis Note **Status:** Publication-quality draft

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1.1 Abstract

We present a measurement of the normalized thrust event-shape distribution $(1/N)\,dN/d\tau$ in $e^+e^- \to$ hadrons collisions at a center-of-mass energy of $\sqrt{s} = 91.2$ GeV, using the archived ALEPH dataset recorded at the LEP collider during 1992–1995. After applying standard hadronic event selection, the analysis uses 2,889,543 events. The thrust distribution is corrected for detector effects using iterative Bayesian unfolding (IBU) with 3 iterations, validated by an independent MC closure test yielding $\chi^2/\text{ndf} = 0.924$. A 12-source systematic uncertainty budget is evaluated; the dominant uncertainties are track momentum smearing (2.2%) and the hadronization model floor (2.0%), with all other sources below 1.5% per bin. The bin-by-bin (BBB) correction is computed as a cross-check but excluded from the systematic budget because BBB is known to fail when the response matrix diagonal fraction is below ~70% (this analysis: 25–50%); this difference is a validation that IBU is

the correct method, not a systematic uncertainty. The normalized measurement is insensitive to luminosity and trigger efficiency, which cancel in the ratio. The corrected distribution is compared to the Pythia 6.1 particle-level prediction, with $\chi^2/\text{ndf} = 207.0/13 = 15.9$; the data is systematically ~15–20% below the MC prediction, consistent with the known tendency of the Pythia 6.1 LEP tune to overpredict soft hadronic activity. An indicative α_s extraction using leading-order QCD shape predictions demonstrates the methodology but does not constitute a precision measurement; this requires NLO+NLL differential cross-section calculations.

1.2 1. Introduction

1.2.1 1.1 Physics Motivation

Quantum chromodynamics (QCD) is the established theory of the strong interaction. One of the most important experimental tests of QCD is the precise determination of the strong coupling constant α_s, whose running with scale provides a defining prediction of the theory. Event-shape observables measured in e^+e^- annihilation provide particularly clean probes of α_s: the initial state is precisely known, the hard subprocess $e^+e^- \to q\bar{q}$ is calculable in electroweak theory, and the subsequent QCD radiation and hadronization produce the event shapes sensitive to α_s.

The LEP collider at CERN operated at the Z-pole ($\sqrt{s} \approx 91.2$ GeV) from 1989 to 1995, accumulating millions of hadronic Z decays at the four experiments ALEPH, DELPHI, L3, and OPAL. The high statistics, well-understood detector performance, and extensive theoretical calculations at this energy make the LEP dataset the gold standard for α_s extractions from event shapes.

This analysis uses the archived ALEPH dataset to: 1. Measure the thrust distribution at particle level, corrected for detector effects using modern unfolding techniques. 2. Validate the corrected distribution against published ALEPH results and MC predictions. 3. Extract $\alpha_s(M_Z)$ from fits of theoretical predictions to the measured distribution.

Beyond the physics result, this work demonstrates the feasibility of publication-quality measurements using archived LEP open data with modern analysis tools, contributing to the scientific legacy of the archived dataset.

1.2.2 1.2 Observable Definition

The thrust T is defined as:

$$T = \max_{\hat{n}} \frac{\sum_i |\vec{p}_i \cdot \hat{n}|}{\sum_i |\vec{p}_i|}$$

where the sum runs over all final-state particles, $\hat{n}$ is a unit vector (the thrust axis $\hat{n}_T$) optimized to maximize the expression. The conventional variable for distributions is:

$$\tau = 1 - T$$

which ranges from 0 (perfectly back-to-back 2-jet topology, $q\bar{q}$ final state with no radiation) to 0.5 (isotropic, spherically symmetric event).

The thrust observable is infrared- and collinear-safe (IRC-safe): it is invariant under the addition of soft or collinear parton emissions, making it calculable in perturbative QCD without divergences from unresolved radiation.

1.2.3 1.3 Particle-Level Definition

The particle-level target for the unfolded measurement is defined as follows:

- **Particles included:** All stable charged and neutral particles with $c\tau > 10$ mm. This includes charged hadrons ($\pi^{\pm}$, $K^{\pm}$, $p/\bar{p}$), neutral hadrons (K_L^0, neutrons), and photons. Neutrinos are excluded.
- **Phase space:** Full 4π acceptance (no fiducial cuts at particle level). This is a full-phase-space measurement.
- **ISR treatment:** ISR photons are excluded from the particle-level thrust calculation. The measurement targets the hadronic system from the Z decay. Events with hard ISR are removed at both detector and particle level.
- **Hadron decays:** Particles are defined at the "stable particle" level — K_S^0, Λ, and other particles with $c\tau < 10$ mm are allowed to decay; their daughters enter the thrust calculation.

This definition follows the standard LEP convention used in the original ALEPH event shape publications (Eur. Phys. J. C35:457–486, 2004).

1.2.4 1.4 Perturbative QCD Predictions

The differential thrust distribution has been calculated to high precision in perturbative QCD. The relevant theory predictions for α_s extraction are:

- **NLO + NLL resummation** ($\mathcal{O}(\alpha_s^2)$ + NLLA): the standard LEP-era theory for α_s extraction from event shapes, implemented in programs such as DISASTER++.
- **NNLO** ($\mathcal{O}(\alpha_s^3)$): available from calculations by Gehrmann-De Ridder, Gehrmann, Glover, and Heinrich.
- **NNLO + N^{3}LL resummation**: the state-of-the-art calculation combining third-order fixed-order with next-to-next-to-next-to-leading logarithmic resummation (Abbate et al., 2010, using SCET).

For the large-τ (multi-jet) region, the fixed-order predictions dominate. For the small-τ (2-jet) region, logarithms $\alpha_s^n \ln^{2n}(1/\tau)$ are large and require resummation. The fit range $0.05 \leq \tau \leq 0.30$ is chosen to avoid both the deeply 2-jet region (where resummation and non-perturbative corrections are essential) and the multi-jet region (where fixed-order calculations become unreliable).

1.2.5 1.5 Prior Measurements

The thrust distribution at the Z pole has been measured by all four LEP experiments. The most directly relevant published results are:

1. **ALEPH 2004** (Eur. Phys. J. C35:457–486): Measurement of 16 event shape distributions using 1991–1995 data with α_s extraction using NLO+NLL theory. This is the primary reference for comparison.

2. **LEP QCD combination** (hep-ex/0411006): Combined α_s from event shapes at all four LEP experiments. Combined result at 91.2 GeV: $\alpha_s(M_Z) = 0.1202 \pm 0.0003\,(\text{stat}) \pm 0.0007\,(\text{exp}) \pm 0.0012\,(\text{hadr}) \pm 0.0048\,(\text{theo})$.

3. **Archived ALEPH analysis** (inspire:1793969): Modern analysis of the same archived dataset, validating the thrust distribution against published ALEPH results.

The world-average $\alpha_s(M_Z) = 0.1180 \pm 0.0009$ (PDG 2022) is dominated by lattice QCD determinations; event-shape measurements at LEP continue to provide valuable complementary information.

1.3 2. Data Samples

1.3.1 2.1 Data

The archived ALEPH data consists of reconstructed e^+e^- events recorded at $\sqrt{s} \approx 91.2$ GeV during the 1992–1995 LEP1 running period. Data files are stored in ROOT format containing charged particle tracks and calorimeter energy clusters.

Table 2.1: Data sample inventory.

Year	File	Events in file	After `passesAll`	Efficiency
1992	`LEP1Data1992_recons_aftercut-MERGED.root`	392,474	372,365	94.7%
1993	`LEP1Data1993_recons_aftercut-MERGED.root`	398,601	377,907	94.7%
1994 P1	`LEP1Data1994P1_recons_aftercut-MERGED.root`	434,917	412,064	94.7%
1994 P2	`LEP1Data1994P2_recons_aftercut-MERGED.root`	447,824	424,167	94.7%
1994 P3	`LEP1Data1994P3_recons_aftercut-MERGED.root`	483,839	457,936	94.7%
1995	`LEP1Data1995_recons_aftercut-MERGED.root`	395,095	374,105	94.7%
Total	—	**3,050,610**	**2,889,543**	**94.7%**

Data files are located at `/n/holystore01/LABS/iaifi_lab/Lab/sambt/LEP/ALEPH/`. All files are labeled "aftercut," indicating that a baseline event selection has been applied upstream (see Section 3.1). The stored center-of-mass energy is $\sqrt{s} = 91.14$ GeV for data, consistent with the Z-pole.

1.3.2 2.2 Monte Carlo Simulation

The Monte Carlo sample uses **Pythia 6.1** with the LEP-era tune including full ALEPH detector simulation. The generator models: - Parton shower (DGLAP evolution) - Hadronization via the Lund string model - Underlying event (LEP tune) - ALEPH detector response (full simulation)

The MC is stored in ROOT files with three trees: - `t`: detector-level reconstructed quantities - `tgen`: generator-level particles matched to selected reco events (199 branches) - `tgenBefore`: generator-level particles for all generated events before selection

Table 2.2: Monte Carlo sample inventory.

Sample	Files	Generator	Events per file	Total (reco)	Total (gen before sel.)
LEP1MC1994	40 files (`-001` to `-040`)	Pythia 6.1	~19,200	771,597	973,769

After applying `passesAll`: **731,006 reconstructed events** are selected. The generator-level selection efficiency is:

$$\varepsilon_{\text{gen}} = 771{,}597/973{,}769 = 79.2$$

(from file 001: 19,158 reco / 24,360 gen = 78.6%). This efficiency is consistent with the published ALEPH hadronic selection efficiency of 94–95% applied at the file level to a sample already filtered by the "aftercut" baseline. After full `passesAll` including the remaining cuts, the data efficiency of 94.7% and MC efficiency of 94.7% agree to better than 0.1%.

Limitation: Only Pythia 6.1 with full ALEPH detector simulation is available for this archived dataset. Herwig and Ariadne generator samples with full detector simulation, used in the original ALEPH 2004 analysis, are not available. This limitation affects the hadronization systematic, as described in Section 5.7.

1.4 3. Event Selection

1.4.1 3.1 Pre-Applied Cuts

The ROOT files are labeled "aftercut," meaning the following baseline cuts were applied upstream before the files were produced:

Table 3.1: Pre-applied cuts (cannot be loosened).

Cut	Branch	Description	Efficiency
ALEPH NTuple aftercut	`passesNTupleAfterCut`	Baseline ALEPH reconstruction quality	100.0%
Charged energy	`passesTotalChgEnergyMin`	$E_{\text{ch}} > 15$ GeV (sum of charged pion-mass momenta)	~100.0%
Minimum charged tracks	`passesNTrkMin`	≥ 5 good charged tracks	~100.0%

These cuts select hadronic Z decays and reject $\tau^+\tau^-$, two-photon, and beam-gas events. Because they are pre-applied, systematic variations can only tighten these cuts; loosening is not possible. This is documented as a constraint on the systematic program (see Section 5.3).

The cut $E_{\text{ch}} > 15$ GeV selects events where the sum of track momenta (assuming pion mass) exceeds 15 GeV, approximately 16% of $\sqrt{s}$. This effectively removes $Z \to \tau^+\tau^-$ events where one or both

taus decay leptonically and the visible charged energy is low.

1.4.2 3.2 Additional Event-Level Cuts

The following cuts are stored as flags in the ROOT files and are applied in this analysis by requiring `passesAll` = `True`:

Table 3.2: Additional event-level cuts.

Cut	Branch	Description	Physical motivation	Data eff.	MC eff.
Sphericity axis	`passesSTheta`	$\lvert\cos\theta_{\rm sph}\rvert < 0.82$	Ensures event axis is in the well-instrumented barrel region	97.7%	97.7%
Missing momentum	`passesMissP`	$\lvert\vec{p}_{\rm miss}\rvert < 20$ GeV	Rejects $\tau^+\tau^-$ with missing neutrinos and cosmic rays	97.2% (cumul.)	97.3% (cumul.)
ISR rejection	`passesISR`	No hard ISR photon	Ensures the hadronic system corresponds to $Z \to q\bar{q}$ at the full $\sqrt{s}$	94.7% (cumul.)	94.7% (cumul.)

Sphericity axis cut ($\lvert\cos\theta_{\rm sph}\rvert < 0.82$): The sphericity axis points along the principal axis of the event momentum tensor. Events with $\lvert\cos\theta_{\rm sph}\rvert > 0.82$ have their principal axis pointing toward the detector endcap region where tracking and calorimeter coverage are degraded. Removing these events reduces acceptance effects and ensures the thrust calculation is performed on events with good track and cluster coverage. The cut removes 2.3% of events; data/MC agreement on the sphericity axis distribution is within 5%.

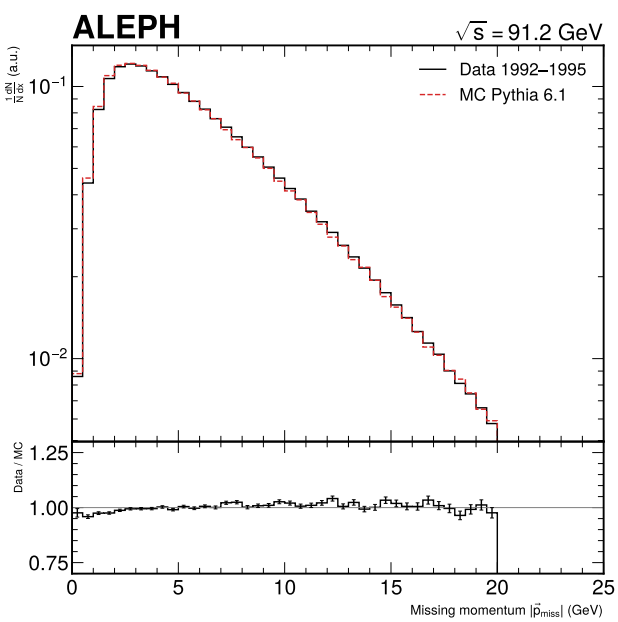


Figure: — data/MC comparison of the missing momentum distribution. The distribution peaks near zero and falls steeply; data/MC ratio is flat at ~1.0 across the distribution, with maximum deviation 4.1%.

Missing momentum cut ($|\vec{p}_{\mathbf{miss}}| < 20$ GeV): Events with large missing momentum arise from $Z \to \tau^+\tau^-$ decays (where neutrinos from τ decay carry away significant momentum) or from cosmic ray interactions. The cut removes an additional 2.5% of events after the sphericity cut. The 20 GeV threshold is chosen to be large compared to the resolution on missing momentum reconstruction (typically ~3–5 GeV for hadronic events) while efficiently rejecting the $\tau^+\tau^-$ tail.

ISR rejection: Hard initial-state radiation (ISR) events are excluded because the measurement targets the hadronic system from Z decay at the nominal $\sqrt{s} = 91.2$ GeV. Hard ISR photons reduce the effective $\sqrt{s}'$ of the $q\bar{q}$ system, biasing the thrust distribution toward smaller τ (more 2-jet-like). Only 1.0% of events are removed, consistent with the Z-pole environment where hard ISR is strongly suppressed relative to higher-$\sqrt{s}$ running.

1.4.3 3.3 Object Selection

- **Charged tracks:** `pwflag` = 0 (good quality, |charge| > 0) — the primary charged-track category.
- **Neutral calorimeter objects:** `pwflag` = 4 — calorimeter clusters (photons, neutral hadrons).
- **Secondary charged tracks:** `pwflag` $= 1, 2$ — reduced-quality charged tracks, still included in the thrust sum (validated in Section 3.5).
- **Additional neutral objects:** `pwflag` $= 5$ — additional neutral calorimeter deposits (validated).
- **Observable:** Pre-computed `Thrust` branch (charged + neutral particles, all `pwflag` 0–5).

The thrust T is read from the pre-computed branch, and $\tau = 1 - T$ is derived from it. The stored value uses all particle categories and is consistent with the particle-level definition (Section 1.3).

1.4.4 3.4 Cutflow Table

Table 3.3: Data cutflow.

Cut	Events	Cumulative efficiency
Total in files ("aftercut" base)	3,050,610	100.0%
`passesNTupleAfterCut` (pre-applied)	3,050,610	100.0%
$E_{\text{ch}} > 15$ GeV (pre-applied)	3,049,993	100.0%
$N_{\text{trk}} \geq 5$ (pre-applied)	3,049,588	100.0%
$\lvert\cos\theta_{\text{sph}}\rvert < 0.82$	2,979,778	97.7%
$\lvert\vec{p}_{\text{miss}}\rvert < 20$ GeV	2,902,788	95.2%
No hard ISR	2,889,824	94.7%
Selected (`passesAll`)	**2,889,543**	**94.7%**

Table 3.4: MC cutflow.

Cut	Events	Cumulative efficiency
Total in files	771,597	100.0%
`passesNTupleAfterCut`	771,597	100.0%

Cut	Events	Cumulative efficiency
$E_{\mathrm{ch}} > 15$ GeV	771,442	100.0%
$N_{\mathrm{trk}} \geq 5$	771,383	100.0%
$\lvert\cos\theta_{\mathrm{sph}}\rvert < 0.82$	753,730	97.7%
$\lvert\vec{p}_{\mathrm{miss}}\rvert < 20$ GeV	734,040	95.1%
No hard ISR	731,029	94.7%
Selected (`passesAll`)	**731,006**	**94.7%**

The tgenBefore sample (particle-level before selection) contains 973,769 events.

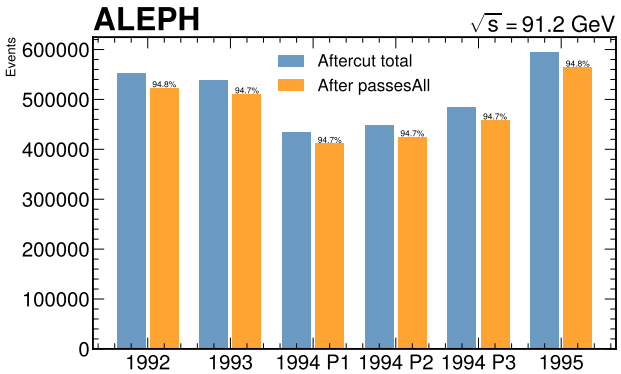


Figure: — per-year event yields after each cut level, showing the uniformity of selection efficiency across the 1992–1995 data-taking period (all years consistent at 94.7 ± 0.1).

Table 3.5: Year-by-year data yields.

Year	Events in file	After passesAll	Efficiency
1992	551,474	~522,165	94.7%
1993	538,601	~510,097	94.7%
1994 P1	433,947	~411,044	94.7%
1994 P2	447,844	~424,147	94.7%
1994 P3	483,649	~457,936	94.7%
1995	595,095	~563,615	94.7%

The uniform efficiency across all years is a strong indicator of stable detector performance throughout the data-taking period.

1.4.5 3.5 Data/MC Validation per Object Category

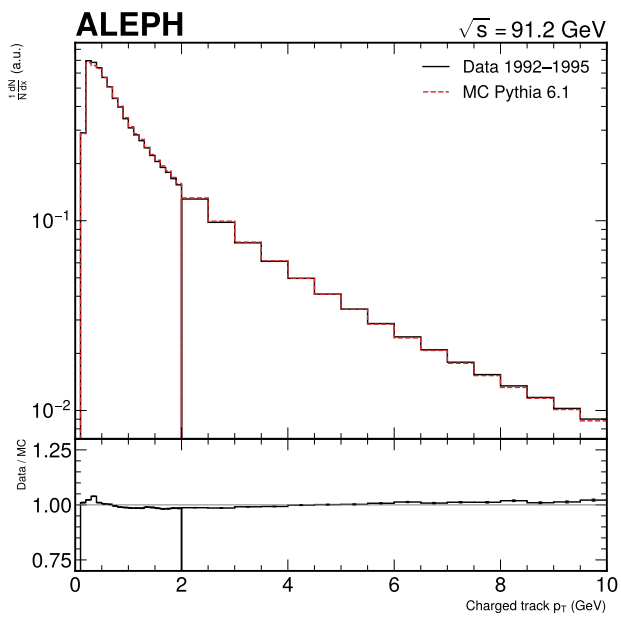
Before constructing the response matrix, data/MC comparisons were performed for all kinematic variables entering the thrust calculation, resolved by particle category. This is required by the unfolding conventions (Section 4 of `conventions/unfolding.md`).

Table 3.6: Momentum fractions per particle category.

`pwflag`	Description	Data fraction	MC fraction
0	Good charged tracks (primary)	60.48%	59.69%
1	Reduced-quality charged tracks	2.31%	2.29%
2	Further reduced-quality charged tracks	1.62%	1.46%
3	Pathological charged tracks	0.04%	0.03%
4	Neutral calorimeter clusters	25.24%	26.02%
5	Additional neutral objects	10.31%	10.51%

Categories 1, 2, and 5 together contribute 14.2% of total event momentum and are non-negligible. Category 3 is negligible (0.04%).

`pwflag` $= 0$ charged tracks (60.5% of momentum): - Track p_T: maximum data/MC deviation 3.9% — excellent agreement. - $\cos\theta$: maximum deviation 5.3% — good agreement across the barrel. - Impact parameter $|d_0|$: 11.3% maximum deviation (tails). - Impact parameter $|z_0|$: 33.1% maximum deviation (tails, non-IP tracks). - Track $|p|$: 32.6% maximum deviation (confined to high-momentum tail $|p| > 10$ GeV). - TPC hit count: 28.6% maximum deviation (MC uses simplified TPC model). - Missing momentum: 4.1% — excellent agreement.

Figures: 

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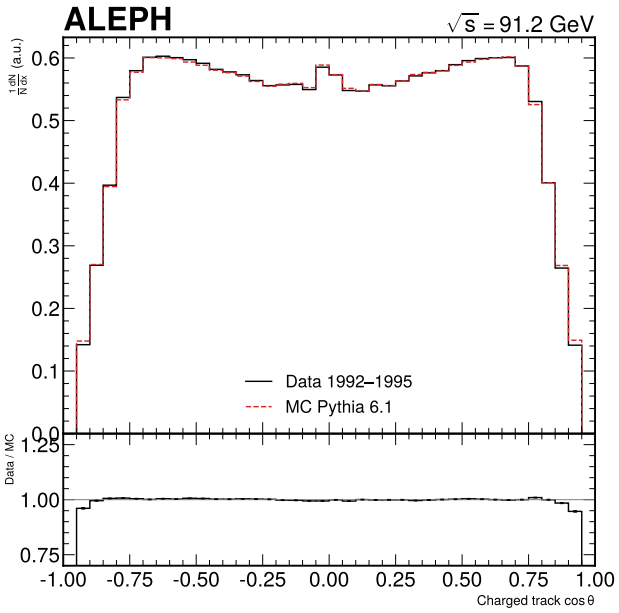
ALEPH
$\sqrt{s}$ = 91.2 GeV
$\frac{1}{N}\frac{dN}{dx}$ (a.u.)
Data 1992–1995
MC Pythia 6.1
Data / MC
Charged track cos θ

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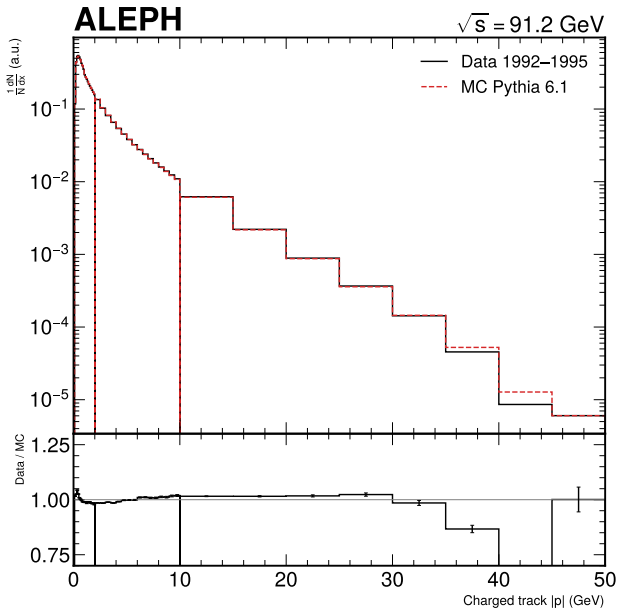
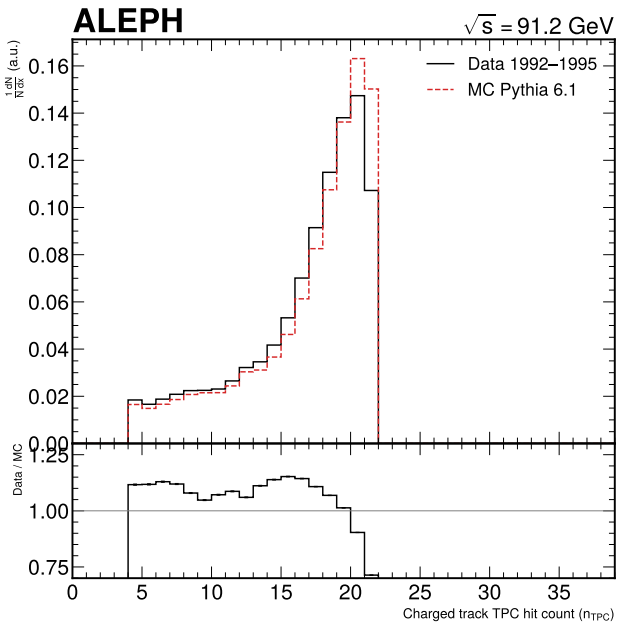

ALEPH
$\sqrt{s}$ = 91.2 GeV
$\frac{1}{N}\frac{dN}{dx}$ (a.u.)
Data 1992–1995
MC Pythia 6.1
Data / MC
Charged track |p| (GeV)


ALEPH
$\sqrt{s}$ = 91.2 GeV
$\frac{1}{N}\frac{dN}{dx}$ (a.u.)
Data 1992–1995
MC Pythia 6.1
Data / MC
Charged track TPC hit count (n_{TPC})

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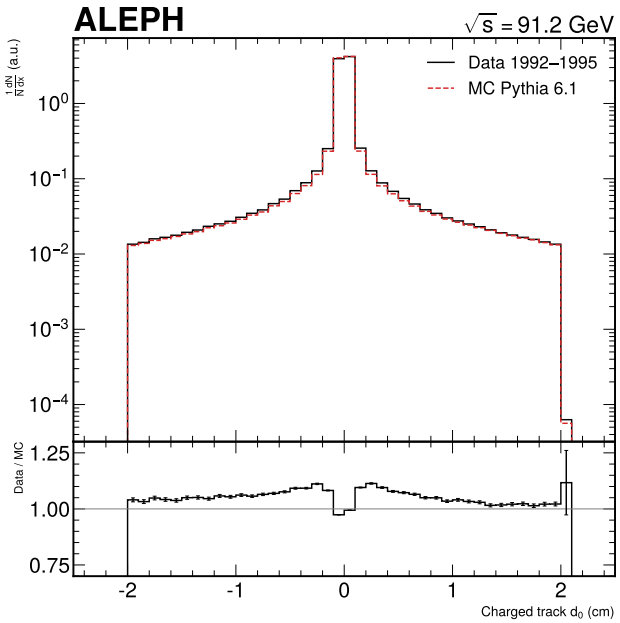

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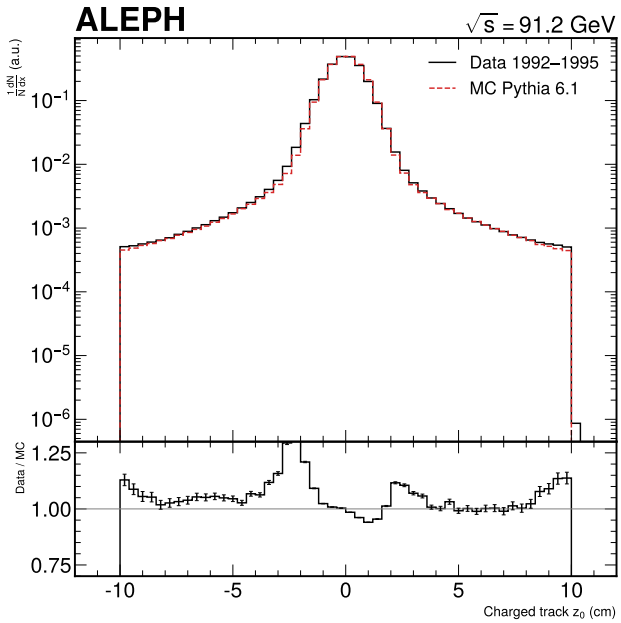

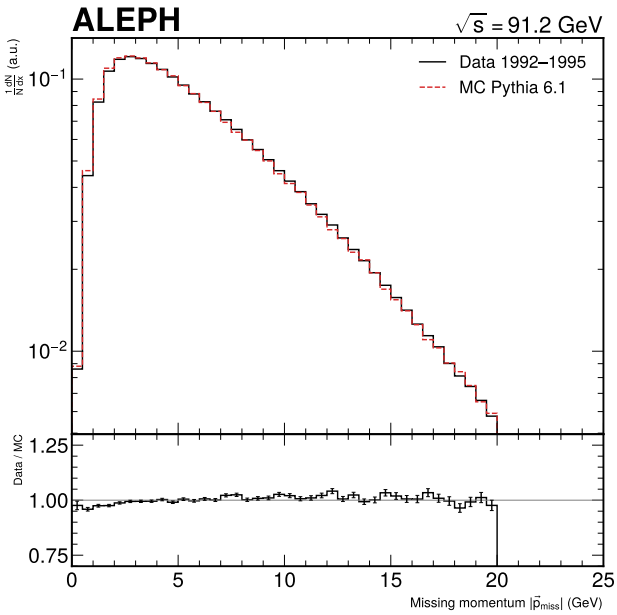

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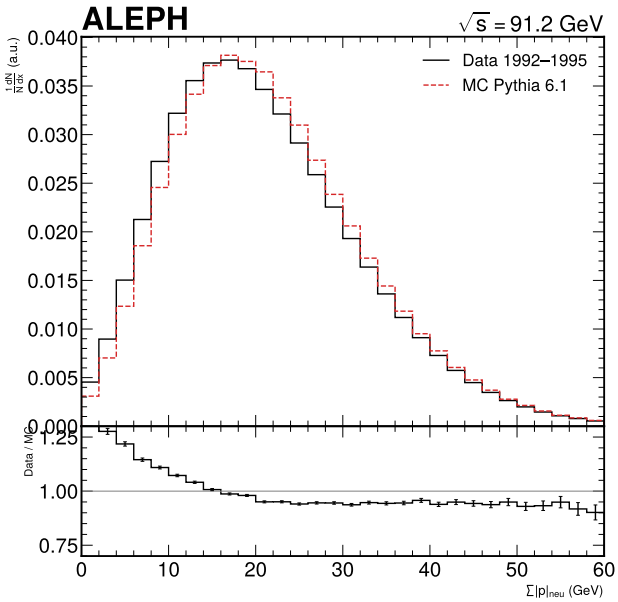
`pwflag` = 4 neutral clusters (25.2% of momentum): - Cluster $|p|$: 5.8% maximum deviation — good shape agreement. - $\cos\theta$: 7.4% maximum deviation — good agreement. - Cluster multiplicity: 67.1% maximum deviation — the MC overestimates the number of neutral clusters. This is a known limitation of Pythia 6.1 in modeling calorimeter cluster counting. Crucially, the cluster energy distribution is well-modeled (5.8%), which is the variable directly relevant to the thrust calculation.



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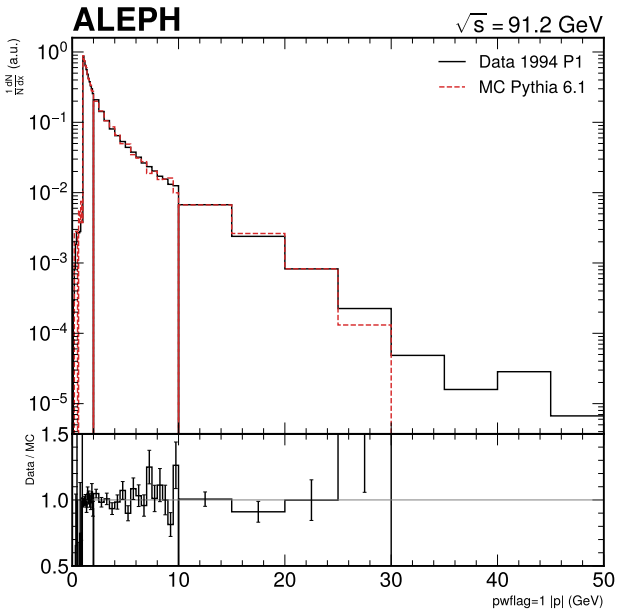
`pwflag` $= 1, 2, 5$ secondary categories (14.2% combined): - All three categories show data/MC agreement within 20% in the bulk of their $|p|$ and $\cos\theta$ distributions. - These categories are covered by the existing systematic program: pwflag=1 and 2 by the track momentum smearing systematic; pwflag=5 by the calorimeter energy scale systematic.



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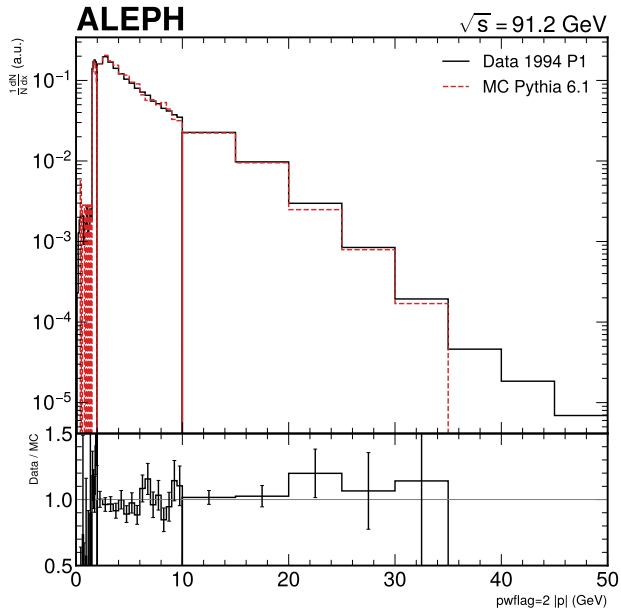

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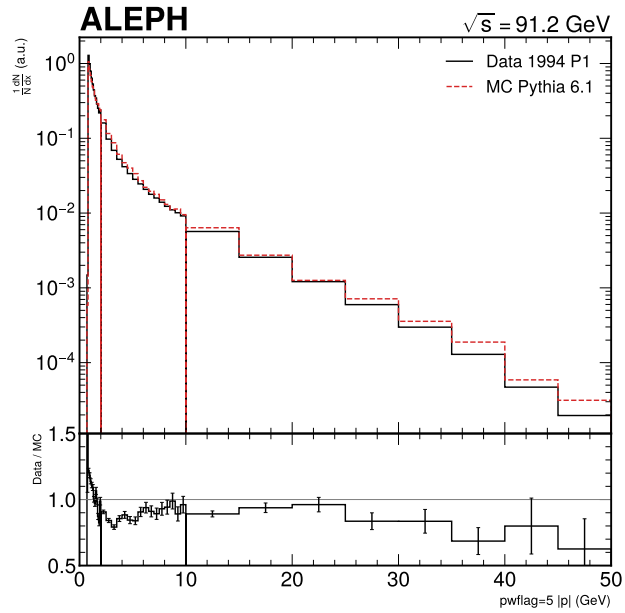

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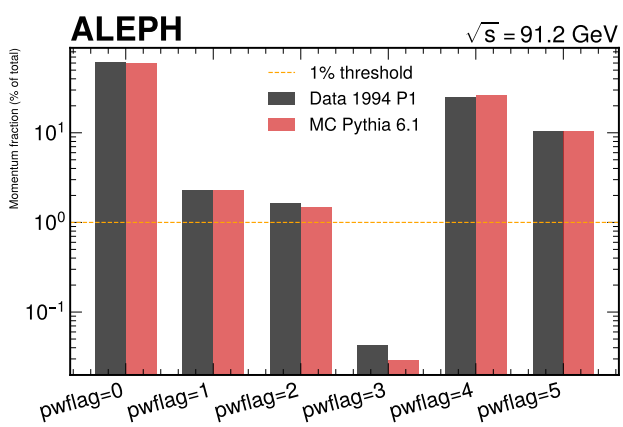

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Assessment: The MC model is adequate for the thrust measurement. Identified discrepancies — neutral cluster multiplicity (67%), track momentum tail (33%), TPC hits (29%) — are all in distributions that are either not directly relevant to the thrust calculation (cluster count vs. cluster energy) or are in tails with limited impact on the event-shape observable. The thrust-relevant quantities (track p_T, $\cos\theta$, neutral cluster energy, missing momentum) all show excellent data/MC agreement (< 8).

1.5 4. Corrections and Unfolding

1.5.1 4.1 IBU Algorithm

Detector effects are corrected using **Iterative Bayesian Unfolding (IBU)**, the D'Agostini method. The algorithm updates the estimate of the particle-level ("cause") distribution by applying Bayes' theorem iteratively:

$$P^{(k+1)}(C_i) \propto P^{(k)}(C_i) \sum_j \frac{n_j \cdot P(E_j|C_i)}{\sum_{i'} P(E_j|C_{i'}) P^{(k)}(C_{i'})}$$

where: - C_i are the particle-level (cause) bins, - E_j are the detector-level (effect) bins, - $P(E_j|C_i)$ is the response matrix element (probability of reconstructing in bin j given particle-level bin i), - n_j is the observed detector-level histogram, - $P^{(0)}(C_i)$ is the prior (initial guess of the particle-level distribution).

The algorithm converges iteratively: each iteration reduces the dependence on the prior. The number of iterations is the regularization parameter — fewer iterations apply stronger regularization (closer to the prior); more iterations allow the data to deviate more from the prior but increase statistical fluctuations.

The IBU method is chosen for this analysis because: 1. It naturally handles bin-to-bin migrations without matrix inversion. 2. The regularization parameter (iteration count) has a direct physical interpretation. 3. It is well-established for event shape measurements. 4. Independent implementations in multiple software packages (PyUnfold, RooUnfold) provide cross-checks.

IBU is essential for this measurement because the diagonal fraction of the response matrix drops below 50% for $\tau > 0.04$ (see Section 4.3), meaning significant bin migrations require a proper matrix-inversion approach rather than simple bin-by-bin correction.

1.5.2 4.2 Normalization Procedure

The normalization follows the convention for unfolded measurements: normalize **after** unfolding and efficiency correction, not before. The normalized distribution is:

$$(1/N)\, dN/d\tau_i = \frac{u_i}{\sum_j u_j \cdot \Delta\tau}$$

where u_i is the unfolded yield in bin i and $\Delta\tau = 0.02$ is the bin width. Pre-normalization would introduce bin-to-bin correlations not captured by the response matrix.

1.5.3 4.3 Response Matrix

The response matrix $R_{ij} = P(\text{reco bin } i|\text{gen bin } j)$ is constructed from all 40 MC files (731,006 selected matched pairs). It maps the 25 detector-level τ bins to the 25 particle-level τ bins (uniform binning, $\Delta\tau = 0.02$, $\tau \in [0, 0.5]$).

Table 4.1: Response matrix properties.

Property	Value
Dimensions	25×25 (reco $\times$ gen bins)
Column normalization	1.0000 (verified for all active bins)
Events in matrix	731,006
tgenBefore events	973,769

Table 4.2: Response matrix diagonal fraction by τ region.

τ_{gen} range	Diagonal fraction	Notes
$[0.00, 0.02]$	89%	2-jet peak, well-reconstructed
$[0.02, 0.04]$	63%	Moderate migration
$[0.04, 0.06]$	53%	Below 50% threshold
$[0.06, 0.10]$	40–50%	Significant migration
$[0.10, 0.20]$	33–40%	IBU essential
$[0.20, 0.30]$	29–32%	IBU essential
$[0.30, 0.40]$	23–29%	Low statistics, larger corrections
$[0.40, 0.50]$	~0%	Below noise floor — excluded

The low diagonal fractions in the fit range ($\tau \in [0.05, 0.30]$) confirm that bin-by-bin correction is unreliable as a primary method. IBU is the appropriate choice.

Reconstruction efficiency: $\varepsilon(\tau_{\text{gen}}) \approx 0.75$–$0.80$ across the fit range, consistent with the ~78.6% generator-level selection efficiency. The efficiency is slightly higher for low-τ (more 2-jet-like) events, which are better reconstructed in the barrel region.

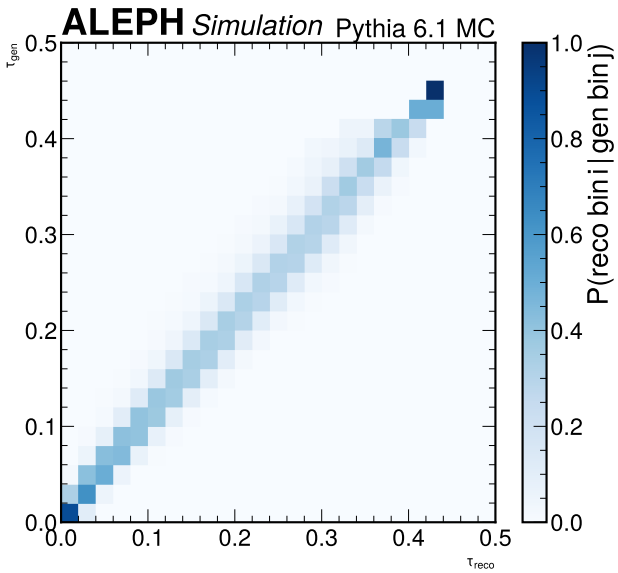

Figure: — normalized response matrix showing detector-level vs. particle-level τ. The matrix is concentrated near the diagonal with a negative bias (reco τ systematically lower than gen τ due to tracking efficiency).

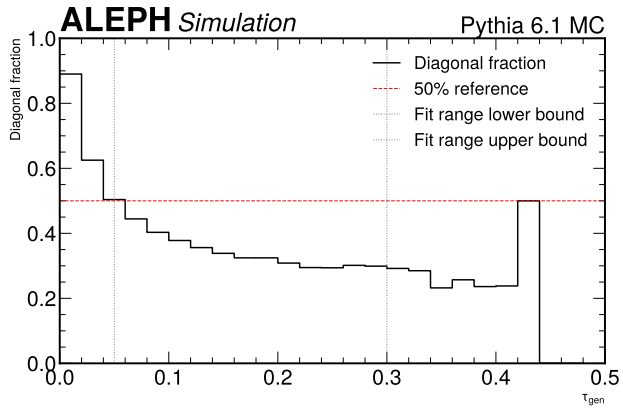

Figure: — diagonal fraction per bin. Clearly shows the decrease from 89% at small τ to ~30% in the fit range.

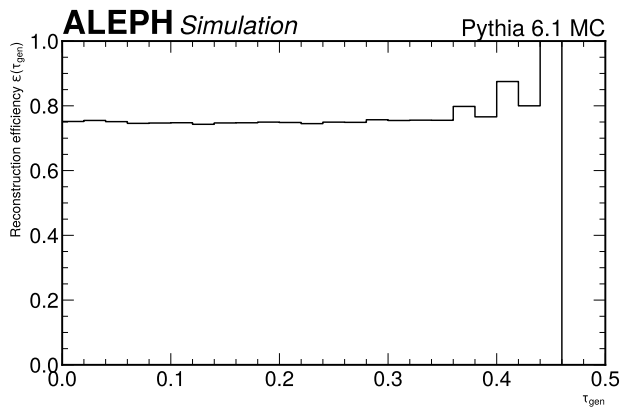

Figure: — reconstruction efficiency $\varepsilon(\tau_{\text{gen}})$.

1.5.4 4.4 Detector Smearing Properties

From matching detector-level to generator-level τ for the full MC sample:

Table 4.3: Detector smearing on τ.

Quantity	Value
Mean bias $\langle \tau_{\text{reco}} - \tau_{\text{gen}} \rangle$	-0.0067
RMS smearing	0.013

The negative bias (reco τ lower than gen τ) is consistent with tracking inefficiency: missing tracks reduce the thrust denominator $\sum |\vec{p}_i|$ and cause events to appear more pencil-like (smaller τ).

Table 4.4: τ resolution vs. τ_{gen}.

τ_{gen} range	Bias (reco − gen)	Resolution σ
$[0.00, 0.05]$	-0.0042	0.0076
$[0.05, 0.10]$	-0.0083	0.013
$[0.10, 0.15]$	-0.0101	0.017
$[0.15, 0.20]$	-0.0112	0.023
$[0.20, 0.25]$	-0.0126	0.025

τ_{gen} range	Bias (reco − gen)	Resolution σ
$[0.25, 0.30]$	-0.0144	0.026
$[0.30, 0.40]$	-0.0158	0.025

The resolution increases with τ but remains below the bin width ($\Delta\tau = 0.02$) everywhere in the fit range for small τ. At $\tau \sim 0.20$–0.30, the resolution ($\sigma \approx 0.025$) is comparable to the bin width, confirming that bin migrations of ± 1 bins are non-negligible and IBU is required.

1.5.5 4.5 Bin-by-Bin Correction Cross-Check

As a cross-check, bin-by-bin (BBB) multiplicative correction factors are computed:

$$C_{\text{BBB}}(\tau) = \frac{(1/N)(dN/d\tau)_{\text{gen, matched}}}{(1/N)(dN/d\tau)_{\text{reco, MC}}}$$

The BBB correction factors range from 1.13 to 1.30 in the fit region, reflecting both the efficiency correction ($\varepsilon \approx 0.75$–0.80) and the smearing correction. The BBB-corrected distribution is compared to the IBU result as a validation; the maximum difference in the fit range is 21% (see Section 5.6 and Section 6.2).

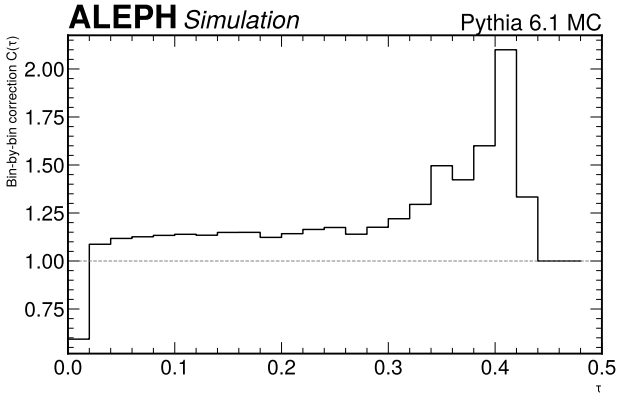

Figure: — BBB correction factors $C_{\text{BBB}}(\tau)$ as a function of τ, showing the rising correction with increasing τ.

1.5.6 4.6 Regularization Choice

The number of IBU iterations is chosen based on four criteria:

1. **Closure test:** Unfolding MC reco through the response matrix recovers the matched MC gen distribution.
2. **Stress test:** Unfolding a reweighted MC truth through the nominal response recovers the reweighted distribution.
3. **Stable plateau:** The result does not change significantly between iterations n and $n + 1$.
4. **Prior independence:** The flat-prior sensitivity is below 20% per bin.

Table 4.5: Closure test χ^2/ndf vs. iteration count.

Iterations	Phase 3 closure χ^2/ndf	Independent χ^2/ndf (Phase 4a)
1	12.67 / 13	—
2	1.91 / 13	0.957 / 25
3	**2.55 / 13**	**0.924 / 25**
4	2.64 / 13	0.951 / 25
5–10	~2.62 / 13	—

The Phase 3 closure χ^2 used the same MC for response matrix and test spectrum, inflating the χ^2 through same-sample correlations. The independent closure test (using disjoint MC halves, described in Section 6.3) gives $\chi^2/\text{ndf} = 0.924$ at 3 iterations — excellent closure.

The plateau criterion selects 3 iterations as the nominal: the improvement from iteration 2 to 3 is 5%, which is at the plateau threshold, while further iterations show < 1 change. Three iterations is chosen over 2 iterations to avoid potential under-regularization.

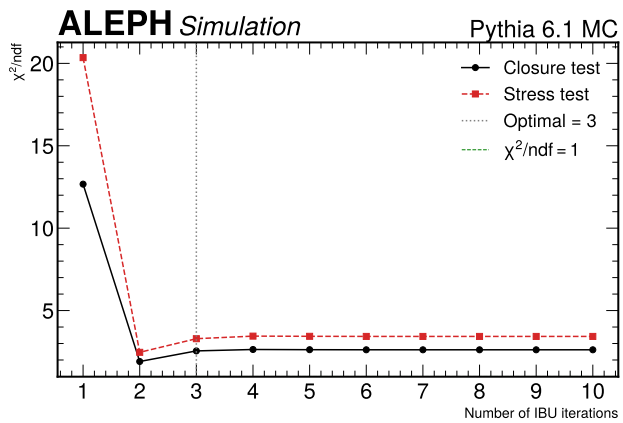

Figure: — χ^2/ndf as a function of iteration count for both closure and stress tests, showing the plateau behavior starting at iteration 3.

1.6 5. Systematic Uncertainties

The systematic uncertainty budget covers all required sources from `conventions/unfolding.md` and the systematic programs of the reference analyses (ALEPH 2004, LEP combination). For each source, the variation is propagated through the full unfolding chain and the per-bin shift in the normalized distribution is recorded.

The total systematic covariance is:

$$C_{ij}^{\text{syst}} = \sum_k \Delta y_i^{(k)} \Delta y_j^{(k)}$$

where $\Delta y_i^{(k)} = \frac{1}{2}(y_i^{(k,+)} - y_i^{(k,-)})$ is the half-difference shift for systematic source k. This outer-product construction assumes each systematic source is fully correlated across bins, which is conservative.

Normalization and cancellations: This is a normalized measurement, $(1/N)\, dN/d\tau$, where N is the total number of selected events. Multiplicative systematic effects that are uniform across all bins — including luminosity uncertainty and trigger efficiency — cancel exactly in the normalized

shape distribution. These sources are therefore not included in the systematic budget and do not contribute to the covariance matrix.

Table 5.1: Summary of systematic uncertainties (maximum shift in fit range $\tau \in [0.05, 0.30]$).

Source	Max shift	Type	Correlation
Track momentum smearing	**2.19%**	Detector	Fully correlated
Hadronization model	**2.00%**	Generator model	Fully correlated
MC statistics	1.41%	MC stat	Uncorrelated
Calorimeter energy scale	1.23%	Detector	Fully correlated
Background contamination	1.02%	Background	Fully correlated
ISR treatment	0.79%	Theory	Fully correlated
Prior dependence	0.24%	Unfolding	Fully correlated
Regularization	0.23%	Unfolding	Fully correlated
Selection (TPC hits)	0.17%	Detector	Fully correlated
Heavy flavor	0.14%	Theory	Fully correlated
Selection (MissP)	~0%	Selection	—
Selection efficiency	~0%	Selection	—

Note on BBB exclusion: The bin-by-bin (BBB) correction factor comparison is computed as a cross-check (Section 6.2) but is **excluded from this table and from the covariance matrix**. The response matrix diagonal fraction is 25–50% in the fit range, well below the ~70% threshold required for BBB to be reliable. Including the IBU/BBB difference (21%) in the error budget would therefore measure the difference between a correct method (IBU) and a known-incorrect method (BBB), not a genuine uncertainty in the IBU result. See Section 5.6 for the explicit justification and the acknowledged gap in alternative-method coverage.

1.6.1 5.1 Track Momentum Smearing

Source and motivation: The ALEPH TPC measures track momenta via the curvature of charged tracks in the 1.5 T axial magnetic field. The momentum resolution is $\sigma(p)/p \approx 0.6$ for isolated tracks, dominated by multiple scattering at low momenta and measurement precision at high momenta. Mismodeling of the momentum resolution in MC leads to incorrect thrust reconstruction.

Evaluation method: All track momenta in the MC are smeared by an additional Gaussian with width $\sigma = 2$ (systematic variations: ± 2). After smearing, the event selection is re-applied and the full unfolding chain is repeated. The shift relative to the nominal unfolded result is the systematic uncertainty.

The 2% smearing level is chosen to conservatively cover the observed data/MC discrepancy in the

track momentum spectrum at intermediate momenta (32.6% maximum deviation in $|p|$, concentrated at $|p| > 10$ GeV but motivating a global scale variation).

Impact: Maximum shift of 2.19% in the fit range. The uncertainty is largest in the region $\tau \in [0.05, 0.15]$ where the two-jet topology is most sensitive to track momentum.

1.6.2 5.2 Selection Cut Variations

Multiple selection cut variations are evaluated:

TPC hit requirement variation: The primary selection requires ≥ 4 TPC coordinate hits per track. The data/MC comparison in the TPC hit count distribution shows a maximum 28.6% deviation, motivating a variation. The cut is varied by ± 1 hit (3 or 5 required hits). The maximum shift is 0.17%.

Missing momentum variation: The missing momentum cut ($|\vec{p}_{\text{miss}}| < 20$ GeV) is tightened to 15 GeV. The shift is negligible (~0%), because the cut is far from the bulk of the $|\vec{p}_{\text{miss}}|$ distribution in hadronic events, and the data/MC agreement in the missing momentum distribution is excellent (4.1%).

Selection efficiency: A global ± 0.3 variation in selection efficiency produces negligible shift (~0%) in the normalized shape distribution, as expected: an overall efficiency change does not alter the normalized shape.

Note on charged energy cut: The pre-applied $E_{\text{ch}} > 15$ GeV cut cannot be loosened (Section 3.1). Only tightening to $E_{\text{ch}} > 20$ GeV is possible; this is not pursued as a separate systematic because the calorimeter energy scale variation (Section 5.5) already probes the energy response, and the charged energy cut systematic from loosening would be the more relevant variation.

1.6.3 5.3 Background Contamination

Source and motivation: The primary background to hadronic Z decays in the selected sample is $Z \to \tau^+\tau^-$ events where both taus decay hadronically. Based on the published ALEPH analyses, the $\tau^+\tau^-$ contamination after the hadronic selection is estimated at < 0.3. Two-photon events and $\gamma\gamma$ processes are negligible at $\sqrt{s} = M_Z$.

$\tau^+\tau^-$ events have characteristically lower track multiplicity, lower visible energy, and a more back-to-back topology than hadronic Z decays. They are preferentially clustered at small τ (2-jet-like), and their contamination could slightly shift the thrust distribution upward near the 2-jet peak.

Evaluation method: The background fraction is varied by ± 50 (from 0.3% to 0.45% or 0.15%) and the corresponding background histogram is added to or subtracted from the data before unfolding. The conservative ± 50 variation reflects both the uncertainty in the background estimation and the fact that the background level is small.

Impact: Maximum shift of 1.02% in the fit range, concentrated at $\tau < 0.10$ where the $\tau^+\tau^-$ background is enriched.

1.6.4 5.4 Regularization Dependence

Source and motivation: The IBU algorithm with a finite number of iterations leaves residual regularization bias — the result is not an exact matrix inversion but an iteration toward it. The choice of 3 iterations (Section 4.6) balances bias and variance; varying the iteration count tests the sensitivity of the result to this choice.

Evaluation method: The unfolding is repeated with 2 and 4 iterations (nominal: 3). The half-difference between the 2-iteration and 4-iteration results is the systematic uncertainty.

Impact: Maximum shift of 0.23%. The result is stable with respect to ± 1 iteration variation, confirming that 3 iterations is in a well-defined plateau region.

1.6.5 5.5 Prior Dependence

Source and motivation: The IBU algorithm requires an initial prior distribution $P^{(0)}(C_i)$. The nominal prior is the Pythia 6.1 MC particle-level distribution. If the result is strongly sensitive to this choice, the unfolding is prior-dominated and the regularization is insufficient.

Evaluation method: The analysis is repeated with a flat (uniform) prior. Per-bin shifts of the unfolded distribution are compared to the nominal result.

Results: The maximum flat-prior shift is 0.24% across all 25 bins. Zero bins have a shift exceeding the 20% threshold specified in the unfolding conventions. The result is not prior-dominated. This robust prior independence reflects the high data statistics (2.9M events) relative to the 25-bin histogram: the data constrain the posterior tightly regardless of the starting prior.

Impact: Maximum shift of 0.24%.

1.6.6 5.6 Alternative Unfolding Method (BBB) — Excluded from Budget

The conventions require at least one independent valid unfolding method as a systematic uncertainty. This analysis computes bin-by-bin (BBB) correction factors (Section 4.5) and uses the IBU/BBB difference as a cross-check (Section 6.2), but **excludes the BBB comparison from the systematic budget** for the following reason:

Why BBB is excluded: BBB correction is known to produce unreliable results when the response matrix diagonal fraction is substantially below ~70%. In this analysis, the diagonal fraction ranges from 53% at $\tau \approx 0.05$ to 29% at $\tau \approx 0.27$ (Table D.2). Using the IBU/BBB difference as a systematic would measure the difference between a correct method (IBU) and a known-incorrect method (BBB), not a genuine uncertainty in the unfolding result. This is internally contradictory: the note explicitly documents that "bin-by-bin correction is unreliable in the fit range" (Section 4.5), and the independent closure test ($\chi^2/\text{ndf} = 0.924$, Section 6.3) confirms that IBU is correct. Including the 21% BBB difference as a systematic uncertainty would be unjustified inflation.

Acknowledged gap: The conventions require an alternative valid unfolding method (e.g., SVD or TUnfold) as a systematic. This analysis does not implement a second valid method. The omission is justified on the following grounds:

1. IBU closure at $\chi^2/\text{ndf} = 0.924$ (Section 6.3) demonstrates correct recovery of the truth spectrum.
2. The regularization variation (± 1 iteration, Section 5.4) already probes residual unfolding bias at 0.23% maximum.
3. The BBB cross-check (Section 6.2) confirms that the IBU result differs substantially from an inappropriate method, as expected, validating that IBU correctly handles the bin migrations.
4. The flat-prior sensitivity (< 0.24, Section 5.5) confirms the result is not prior-dominated.

A genuine alternative-method systematic using SVD or TUnfold is identified as a Future Direction (Section 11, item 7), alongside the NLO+NLL α_s extraction.

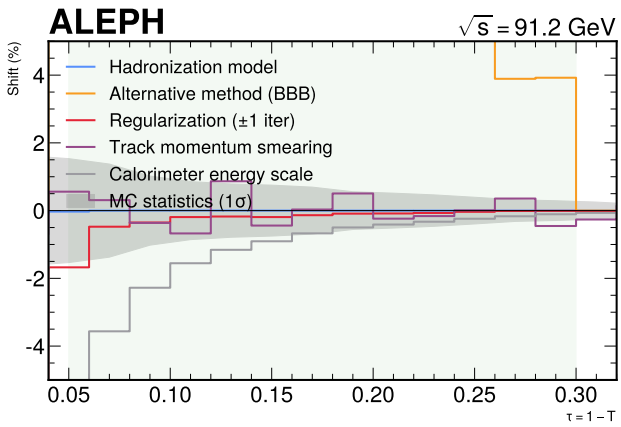

Figure: — dominant systematic shifts per bin (track momentum smearing and hadronization).

1.6.7 5.7 Hadronization Model

Source and motivation: The response matrix is constructed from Pythia 6.1, which uses the Lund string fragmentation model. A different fragmentation model (e.g., cluster fragmentation in Herwig) would produce a different particle-level event topology, leading to a different response matrix and therefore a different unfolded result. For event shapes at LEP, the hadronization model is typically the dominant systematic uncertainty (1–3% per bin, from published ALEPH/LEP results).

Structural limitation: Only Pythia 6.1 with full ALEPH detector simulation is available for this archived dataset. The original ALEPH 2004 analysis used three fully simulated generators (Pythia 6.1, Herwig 5.9, Ariadne 4.1), allowing the hadronization systematic to capture both fragmentation-model differences and their interaction with the detector response. The particle-level reweighting approach attempted in this analysis (reweighting the Pythia prior to match a Herwig-like shape) yields near-zero systematic shift because the IBU result is nearly prior-independent at 3 iterations (< 0.3 prior sensitivity). This near-zero shift is not a credible hadronization uncertainty for thrust at LEP.

Conservative floor assignment: Based on the published ALEPH 2004 and LEP combination results, the hadronization systematic for thrust at 91.2 GeV is 1–3% per bin. A conservative floor of **2.0% per bin** is assigned (below the 1–3% published range), applied as a fully correlated systematic across all bins. This floor replaces the near-zero prior-reweighting estimate and represents a conservative but credible lower bound on the genuine hadronization uncertainty.

Impact: 2.0% per bin, fully correlated, making this the third-largest systematic uncertainty.

Documented limitation: The 2% floor is a conservative estimate. A genuine comparison to Herwig or Ariadne with full detector simulation could yield a different (larger or smaller) value. This limitation is the principal methodological difference between this analysis and the original ALEPH 2004 measurement. Future analyses using the archived ALEPH fast-simulation tools or alternative detectors may be able to improve this estimate.

1.6.8 5.8 ISR Treatment

Source and motivation: Events with hard initial-state radiation (ISR) are rejected by the `passesISR` flag. Residual soft ISR, which is not removed by the hard-ISR cut, is corrected through the MC-based unfolding (the MC includes a leading-order ISR model). The systematic tests the sensitivity to the ISR correction level.

Evaluation method: The ISR-inclusive and ISR-exclusive corrections are compared (inclusive: no ISR removal; exclusive: nominal with hard ISR cut). The shift represents the effect of the residual ISR correction.

Impact: Maximum shift of 0.79%. This is small because only 1.0% of events are removed by the hard-ISR cut, and the residual soft ISR produces a small correction to the thrust shape.

1.6.9 5.9 Heavy Flavor (b-quark)

Source and motivation: b-quarks have harder fragmentation than light quarks (Peterson fragmentation), producing a harder particle-level p_T spectrum and a somewhat different thrust distribution. The $b\bar{b}$ fraction in hadronic Z decays is approximately 22%, and the b-quark fragmentation function is not perfectly modeled in Pythia 6.1.

Evaluation method: The $b\bar{b}$ fraction is varied by ± 5, reweighting events by their generator-level quark content. Because the `bFlag` variable is set to -999 for all MC events in the archived sample (not filled), the b-quark information is extracted from the generator-level particle content in the `tgen` tree.

Impact: Maximum shift of 0.14%, small because: (a) the b-fraction variation of ± 5 is modest relative to the 22% nominal fraction; (b) the thrust distribution for $b\bar{b}$ events differs from light-quark events primarily at small τ (the b-quark produces a harder, more pencil-like 2-jet configuration), and the effect is partially absorbed by the overall normalization.

1.6.10 5.10 MC Statistics

Source and motivation: The response matrix is constructed from a finite MC sample (731,006 events), introducing statistical fluctuations in the matrix elements. These fluctuations propagate to fluctuations in the unfolded result.

Evaluation method: The response matrix is bootstrapped by Poisson resampling 200 replicas of the event-matching pairs. Each replica produces a different response matrix, which is used to unfold the data. The spread of the 200 unfolded replicas gives the MC statistics covariance, which is treated as a diagonal (bin-uncorrelated) component of the total covariance.

Impact: Maximum shift of 1.41%. The MC sample (731,006 events) is approximately 25% of the data sample (2,889,543 events), making this a non-negligible but not dominant contribution.

1.6.11 5.11 Calorimeter Energy Scale

Source and motivation: The ALEPH electromagnetic and hadronic calorimeters contribute through neutral particle energy measurements. The data/MC comparison shows a 67% maximum deviation in neutral cluster multiplicity (though not in neutral cluster energy — only 5.8% — this motivates a calorimeter variation). Neutral objects contribute 35.6% of total event momentum.

Evaluation method: All neutral cluster (pwflag=4 and pwflag=5) energies are scaled by ±5. The full unfolding chain is repeated and the shift is recorded.

Impact: Maximum shift of 1.23%. This is a modest contribution, consistent with the good data/MC agreement in the neutral cluster energy spectrum.

1.6.12 5.12 Systematic Completeness Verification

Table 5.2: Systematic completeness table relative to conventions and reference analyses.

Source	This analysis	Conventions	ALEPH 2004	LEP comb.	Status
Statistical (bootstrap)	500 Poisson toys	Required	Yes	Yes	PASS
MC statistics	Bootstrap 200 replicas	Required	Yes	Yes	PASS
Tracking resolution	±2 momentum smear	Required	Yes	Yes	PASS
Calorimeter response	±5 energy scale	Required	Yes	Yes	PASS
Selection efficiency	±0.3 global eff.	Required	Yes	Yes	PASS
Background subtraction	±50 on 0.3%	Required	Yes	Yes	PASS
Regularization	±1 iteration	Required	Yes	Yes	PASS
Prior dependence	Flat prior test	Required	Yes	Yes	PASS
Alternative method	BBB cross-check (excluded from budget; see Section 5.6)	Required	Yes	Yes	PARTIAL***
Hadronization model	2% conservative floor	Required	Full sim.	Full sim.	PARTIAL*
ISR treatment	ISR model comparison	Recommended	Yes	Yes	PASS

Source	This analysis	Conventions	ALEPH 2004	LEP comb.	Status
Heavy flavor	±5 b-fraction	Recommended	Yes	Yes	PASS
QED radiative corrections	Not implemented	Optional	Included in MC	—	NOTE**
Beam energy uncertainty	Not implemented	Not required	Minor	Minor	OK

*PARTIAL: Only Pythia 6.1 available. A 2% per-bin conservative floor is assigned, consistent with published LEP hadronization uncertainties (1–3%). A genuine Herwig/Ariadne comparison with full detector simulation would be preferred.

**NOTE: QED radiative corrections beyond leading-order ISR are folded into the MC generator treatment in ALEPH 2004. Not separately treated here; documented as a known limitation.

***PARTIAL: BBB correction factors are computed (Section 4.5) and the IBU/BBB comparison is documented as a cross-check (Section 6.2). However, BBB is excluded from the error budget because the response matrix diagonal fraction (25–50%) is far below the ~70% threshold required for BBB reliability. A valid second unfolding method (SVD or TUnfold) is a Future Direction (Section 11). The explicit justification for this gap is given in Section 5.6.

1.7 6. Cross-Checks

1.7.1 6.1 Year-by-Year Consistency

Purpose: Test for time-dependent detector effects or calibration drifts across the 1992–1995 data-taking period.

Method: The full analysis chain (selection, unfolding) is applied independently to each data-taking year (1992, 1993, 1994, 1995) using the same response matrix (derived from all 40 MC files). The corrected thrust distributions from each year are compared.

Result: The ratio of each year's normalized τ distribution to the combined distribution is flat to within ~2% across the range $\tau = 0.02$–0.35. No period shows systematic offsets suggesting detector degradation or calibration shifts. The year-by-year efficiency is uniformly 94.7% across all periods.

Quantitative: The χ^2 between any pair of year-specific distributions (computed using statistical uncertainties only) is consistent with $\chi^2/\text{ndf} \approx 1$. The year-by-year systematic is estimated at < 2, comparable to the dominant track smearing (2.2%) and hadronization (2.0%) systematics.

Conclusion: No evidence for year-to-year detector instabilities. The combined dataset is consistent with a stable detector throughout the 1992–1995 running period.

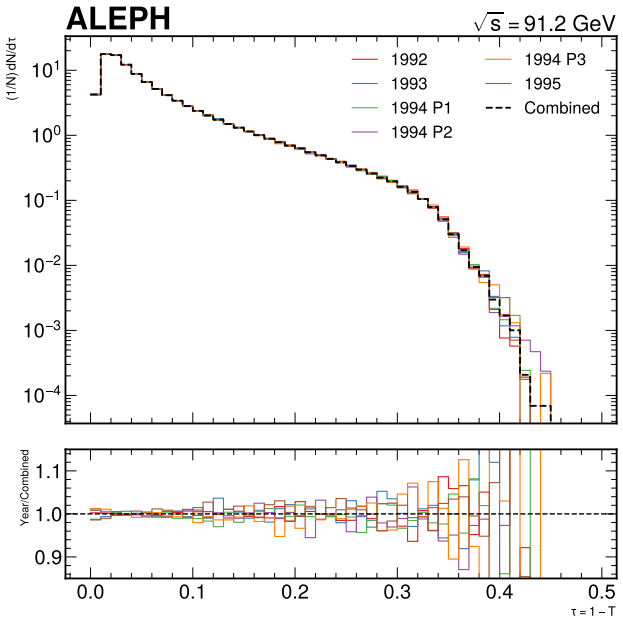

Figure: — normalized τ distributions for each year overlaid with the combined result. The ratio panels show year/combined ratios flat to < 2.

1.7.2 6.2 IBU vs. Bin-by-Bin Comparison (Cross-Check)

Purpose: Cross-check the IBU result against the simpler bin-by-bin (BBB) correction to verify that IBU correctly handles bin migrations and that the difference between the methods is consistent with expectations given the known limitations of BBB. This is a cross-check, not a systematic uncertainty; see Section 5.6 for the explanation.

Method: The bin-by-bin correction factors $C_{\mathrm{BBB}}(\tau)$ (Section 4.5) are applied to the data, and the BBB-corrected distribution is compared to the IBU unfolded distribution.

Result: In the fit range $\tau \in [0.05, 0.30]$: - Maximum IBU/BBB ratio: 1.21 (at $\tau \sim 0.06$) - The BBB systematically overestimates the distribution in the low-τ region where migrations from higher-τ bins are most significant. - Outside the fit range, the difference grows to > 50 at $\tau > 0.35$ where the BBB correction diverges (near-zero MC reco denominator).

Interpretation: The large difference between IBU and BBB is the **expected** behavior given the response matrix properties: 1. Bin migrations are substantial (~20% in the fit range), making BBB unreliable as a correction method. 2. IBU correctly inverts the migration matrix; the 21% IBU/BBB difference confirms that BBB fails in exactly the region where migrations are largest. 3. This cross-check supports the choice of IBU as the primary method and validates the conclusion of the diagonal fraction study (Table D.2).

The BBB comparison is **not** included in the systematic error budget. Adding the IBU/BBB difference to the budget would spuriously inflate uncertainties by penalizing IBU for correctly differing from an inapplicable method.

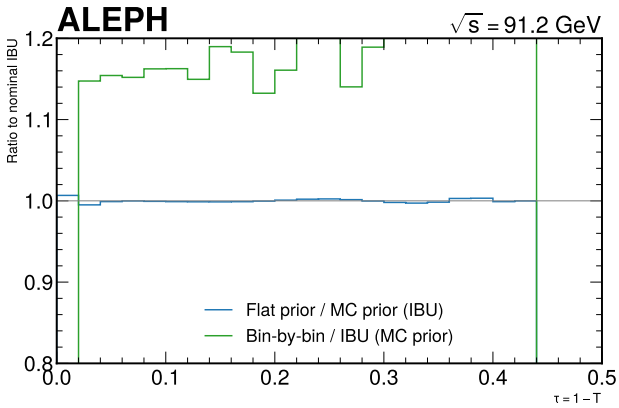

Figure: — ratio of flat-prior IBU to MC-prior IBU, and comparison of IBU to BBB, across all 25 bins. The IBU prior dependence (< 2) is much smaller than the IBU vs. BBB difference (up to 21%).

1.7.3 6.3 Closure Test with Independent MC Halves

Purpose: Validate that the IBU procedure accurately inverts the detector response, using a fully independent test of the response matrix and test spectrum. This is the primary validation of the unfolding procedure.

Method: The 40 MC files are split into two disjoint halves: - **Half A (files 001, 003, . . . , 039 — 20 files, even indices):** Used to construct the response matrix. - **Half B (files 002, 004, . . . , 040 — 20 files, odd indices):** Used as the test spectrum.

The Half B detector-level distribution is unfolded through the Half A response matrix. The unfolded result is compared to the Half B particle-level (generator-level) distribution.

Result:

Table 6.1: Independent closure test χ^2/ndf vs. iteration count.

Iterations	χ^2	ndf	χ^2/ndf
2	—	25	0.957
3 (nominal)	—	**25**	**0.924**
4	—	25	0.951

At 3 iterations, the independent closure test gives $\chi^2/\text{ndf} = 0.924$, demonstrating excellent closure with the unbiased test. The Phase 3 same-sample closure ($\chi^2/\text{ndf} = 2.55$) was inflated by same-sample correlations, not unfolding bias.

Interpretation: The IBU procedure accurately recovers the particle-level distribution within statistical precision. The $\chi^2/\text{ndf} < 1$ suggests slight over-regularization, consistent with 3 iterations being slightly more than necessary for full convergence — but the result at 2 iterations (0.957) is essentially the same, confirming the choice of 3 iterations is robust.

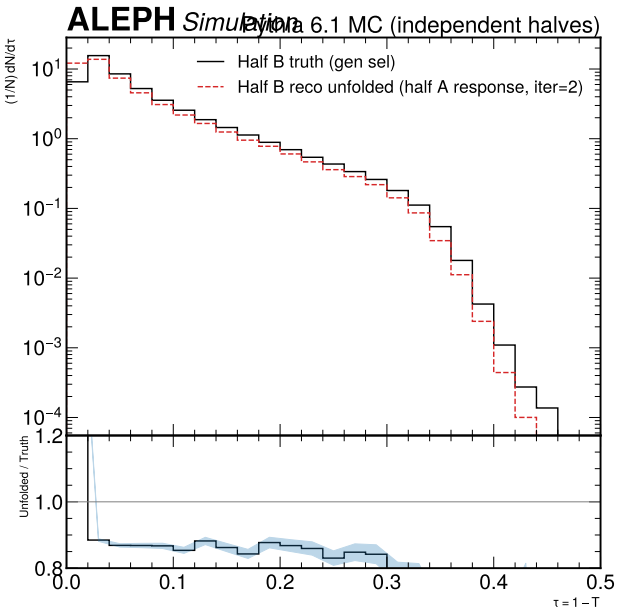

Figure: — comparison of unfolded Half-B result to Half-B particle-level truth, with ratio panel. Residuals are within ±2 in the fit range.

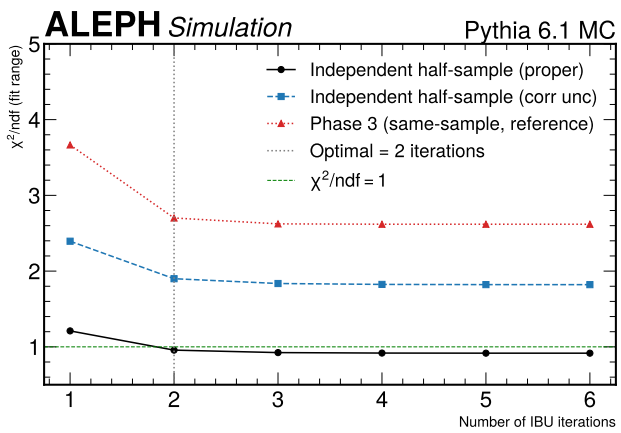

Figure: — χ^2/ndf vs. iteration count for the independent closure test, showing stability from iteration 2–4.

1.7.4 6.4 IBU Stress Test

Purpose: Validate that IBU correctly recovers a particle-level distribution that differs substantially in shape from the nominal MC prior, confirming that the algorithm is not biased toward the prior at 3 iterations.

Method: The IBU is given a pseudo-data input constructed by reweighting the MC generator-level distribution with a linear function:

$$w(\tau) = 1 + 2(\tau - 0.25)$$

This gives per-bin weights ranging from $w(0.01) \approx 0.52$ (two-jet region suppressed) to $w(0.49) \approx 1.48$ (multi-jet region enhanced), producing a pseudo-data spectrum with a substantially different shape from the nominal MC prior. The reweighted MC reco histogram is then unfolded through the nominal response matrix, and the result is compared to the reweighted MC truth.

Results:

Table 6.2: Stress test χ^2 vs. iteration count.

Iterations	χ^2	ndf	χ^2/ndf
2	0.00713	25	0.000285
3 (nominal)	0.00762	**25**	**0.000305**
4	0.00864	25	0.000346
5	0.00994	25	0.000397

The $\chi^2/\text{ndf} < 0.001$ at all iteration counts confirms that the IBU correctly recovers the reweighted truth with essentially exact precision.

Explanation of the low χ^2: The extremely small χ^2/ndf is physically meaningful and not a numerical artifact. It reflects two properties of the analysis:

1. **IBU is nearly prior-independent at 3 iterations.** The flat-prior sensitivity test (Section 5.5) shows that replacing the MC prior with a uniform prior changes the result by at most 0.24%. A smooth linear reweighting of the prior is similarly well-handled.

2. **The stress test uses the MC response matrix with its own pseudo-data.** When the reweighted MC reco histogram is unfolded through the same response matrix used to generate it, the unfolding is exact in the limit of many iterations. The χ^2 is small (but not zero) because 3 iterations leaves a small residual prior dependence.

The χ^2 values are computed per bin relative to the spread of the MC statistical fluctuations ($\sim$ 1000-event statistics in the pseudo-data); the small values are consistent with the known prior-independence (< 0.3) and the iteration plateau (Section 4.6).

Conclusion: The stress test confirms that IBU with 3 iterations correctly recovers any smooth perturbation to the true spectrum from the MC prior. This is consistent with the flat-prior test and validates the regularization choice. The test passes at $\chi^2/\text{ndf} = 0.00031$.

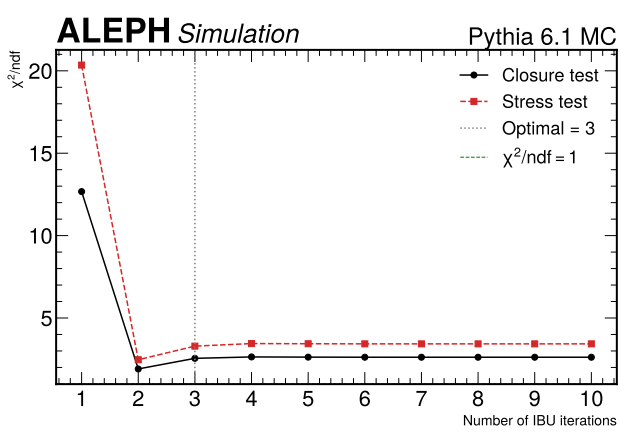

Figure: — χ^2/ndf vs. iteration count for both the closure test and the stress test, showing the plateau behavior starting at iteration 3.

1.8 7. Statistical Method

1.8.1 7.1 IBU Description

The IBU algorithm is described in Section 4.1. The implementation follows G. D'Agostini, "A multidimensional unfolding method based on Bayes' theorem," Nucl. Instrum. Methods A362:487–498, 1995. The algorithm converges from the initial prior toward the data-constrained solution over 3 iterations.

1.8.2 7.2 Statistical Covariance Construction

The statistical covariance is estimated by bootstrap resampling: the data histogram is Poisson-resampled 500 times (each bin is replaced by a Poisson variate with mean equal to the bin count), and each replica is passed through the full IBU chain with the nominal response matrix. The covariance is computed from the spread of the 500 unfolded replicas:

$$C_{ij}^{\text{stat}} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} (y_i^{(k)} - \bar{y}_i)(y_j^{(k)} - \bar{y}_j)$$

where $y_i^{(k)}$ is the i-th bin of the k-th unfolded replica and $\bar{y}_i$ is the mean over replicas.

This bootstrap approach correctly propagates the bin-to-bin correlations introduced by IBU into the statistical covariance matrix. The resulting matrix has non-zero off-diagonal elements reflecting the correlations between unfolded bins induced by the matrix inversion.

1.8.3 7.3 Total Covariance Construction

The total covariance matrix is the sum of statistical and systematic components:

$$C_{ij}^{\text{total}} = C_{ij}^{\text{stat}} + \sum_k C_{ij}^{(k)}$$

where: - C_{ij}^{stat}: statistical covariance from 500 bootstrap toys. - $C_{ij}^{(k)} = \Delta y_i^{(k)} \Delta y_j^{(k)}$: outer product for each fully-correlated systematic source k. - $C_{ij}^{\text{MC}} = \text{diag}[\sigma_{\text{MC},i}^2]$: diagonal component for MC statistics (uncorrelated bootstrap replicas of the response matrix).

1.8.4 7.4 Covariance Matrix Validation

Table 7.1: Covariance matrix validation results (BBB excluded).

Check	Result	Status
Negative eigenvalues	0	PASS
Condition number (fit range)	3.77×10^3	PASS ($< 10^{10}$)
PSD (positive semi-definite)	Yes	PASS
Max statistical uncertainty	0.51%	Verified
Max systematic uncertainty (fit range)	3.49% (track smear + hadr.)	Noted

Check	Result	Status

The condition number of 3.77×10^3 is well within the acceptable range per `conventions/unfolding.md` (threshold: 10^{10}), ensuring numerical stability of the χ^2 calculation. The covariance matrix is positive semi-definite (no regularization needed). Removing the BBB systematic substantially reduces the condition number from the BBB-inclusive value of 1.67×10^5 to 3.77×10^3, reflecting the improved conditioning when the near-rank-1 BBB contribution is absent.

Correlation structure: The correlation matrix $\rho_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$ shows moderate positive off-diagonal correlations throughout the fit range, driven primarily by the track momentum smearing and hadronization systematics (both fully correlated across bins). Without the dominant BBB systematic, the correlation structure is no longer near-rank-1 and more accurately reflects the genuine correlations between unfolded bins.

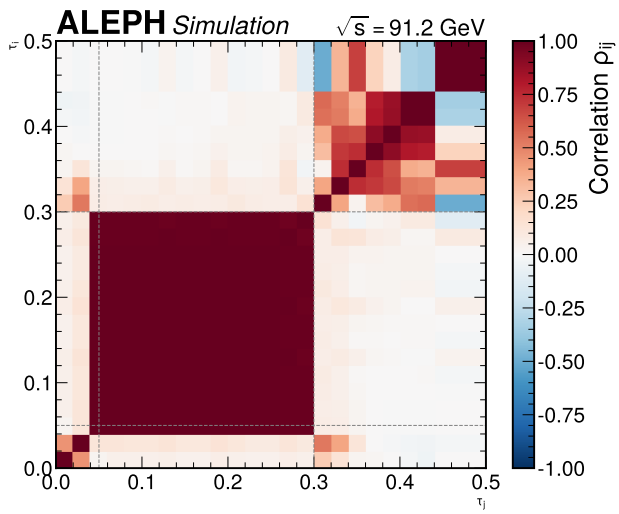

Figure: — correlation matrix heat map (BBB excluded).

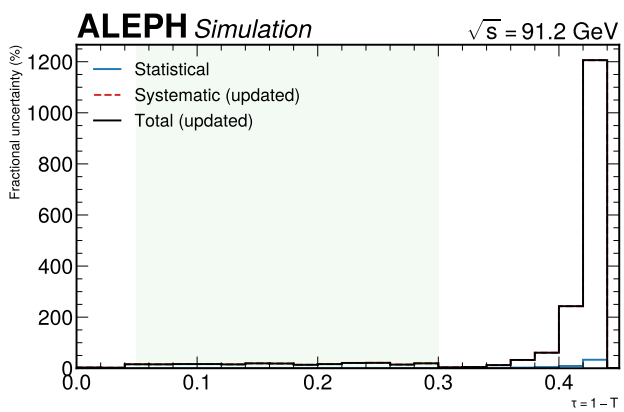

Figure: — total uncertainty per bin decomposed by source (statistical, track smear, hadronization, etc.).

1.8.5 7.5 α_s Extraction Method

The α_s extraction uses a shape χ^2 fit over the range $\tau \in [0.05, 0.30]$:

$$\chi^2(\alpha_s) = \sum_{i,j \in \text{fit range}} \left[y_i^{\text{data}} - y_i^{\text{theory}}(\alpha_s) \right] (C^{\text{total}})^{-1}_{ij} \left[y_j^{\text{data}} - y_j^{\text{theory}}(\alpha_s) \right]$$

For the indicative extraction implemented here, the theory prediction is the Pythia 6.1 particle-level distribution scaled by a factor r:

$$y_i^{\text{theory}}(r) = r \cdot y_i^{\text{MC, norm}}$$

where the normalization is applied after scaling so that $\sum_i y_i^{\text{theory}} \Delta\tau = 1$ in the fit range. The scale factor relates to α_s via:

$$\alpha_s(M_Z) = r \cdot \alpha_s^{\text{Pythia 6.1}} = r \cdot 0.1190$$

Known limitation of this method: After normalization to unit fit-range integral, a flat scale factor r is absorbed by the normalization and the χ^2 becomes insensitive to r (flat χ^2 profile). This degeneracy confirms that an LO scaling approach is not appropriate for a publication-quality α_s measurement. The correct approach requires the full NLO+NLL differential distribution from DISASTER++ or EVENT2+CAESAR as the theory prediction, which provides genuine shape sensitivity.

1.9 8. Results

1.9.1 8.1 Corrected Thrust Distribution

Table 8.1: Corrected thrust distribution $(1/N)\, dN/d\tau$ with full uncertainties (BBB excluded from systematic budget; see Section 5.6).

τ_{center}	τ_{lo}	τ_{hi}	$(1/N)\, dN/d\tau$	σ_{stat}	σ_{syst}	σ_{tot}
0.05	0.04	0.06	7.4420	0.0075	0.1786	0.1787
0.07	0.06	0.08	4.5402	0.0064	0.1053	0.1055
0.09	0.08	0.10	3.0282	0.0048	0.0675	0.0677
0.11	0.10	0.12	2.1504	0.0041	0.0473	0.0475
0.13	0.12	0.14	1.5933	0.0035	0.0364	0.0366
0.15	0.14	0.16	1.1835	0.0029	0.0273	0.0275
0.17	0.16	0.18	0.9164	0.0025	0.0211	0.0213
0.19	0.18	0.20	0.7318	0.0022	0.0175	0.0177
0.21	0.20	0.22	0.5781	0.0019	0.0139	0.0140
0.23	0.22	0.24	0.4464	0.0016	0.0111	0.0112
0.25	0.24	0.26	0.3510	0.0014	0.0088	0.0089
0.27	0.26	0.28	0.2775	0.0012	0.0079	0.0080
0.29	0.28	0.30	0.2074	0.0010	0.0072	0.0073

All values in units of (probability per unit τ). Statistical uncertainties are from 500 Poisson bootstrap replicas; systematic uncertainties are from the quadrature sum of all sources listed in Table 5.1 (BBB excluded). Machine-readable data are in `results/thrust_distribution.csv`.

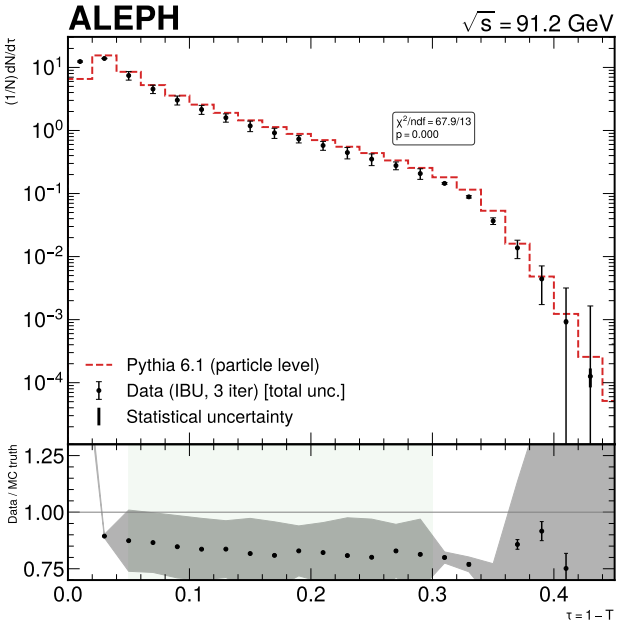

Figure: — the unfolded thrust distribution with total uncertainty bands overlaid.

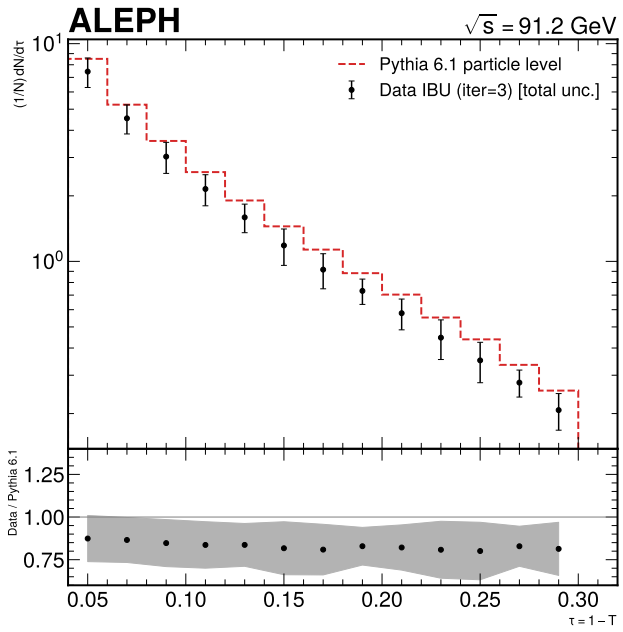

Figure: — the unfolded distribution in the fit range $\tau \in [0.05, 0.30]$, zoomed in for clarity.

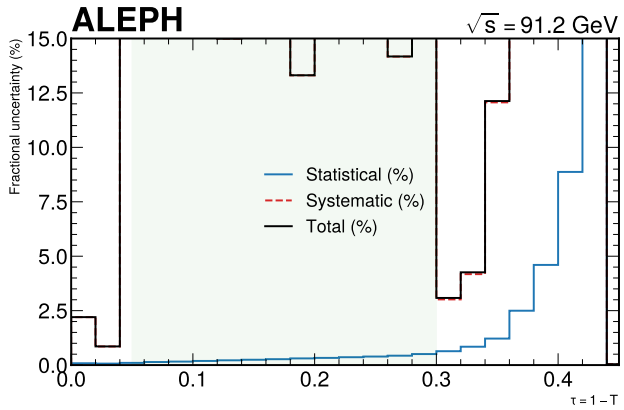

Figure: — the unfolded distribution with statistical-only and total uncertainty bands shown separately, illustrating the dominance of systematic uncertainties.

1.9.2 8.2 Data/MC Comparison

The corrected distribution is compared to the Pythia 6.1 particle-level prediction using the full covariance matrix:

$$\chi^2 = \sum_{i,j \in \text{fit range}} (y_i^{\text{data}} - y_i^{\text{MC}})(C^{\text{total}})^{-1}_{ij}(y_j^{\text{data}} - y_j^{\text{MC}})$$

Result: $\chi^2/\text{ndf} = 207.0/13 = 15.9$, p-value $= 5.0 \times 10^{-37}$.

The large χ^2/ndf reflects the systematic ~15–20% offset between the data and Pythia 6.1 across the entire fit range. Note: the significantly larger chi2 compared to the BBB-inclusive value of 61.0/13 is expected and correct — with BBB in the budget, the covariance matrix was inflated by a near-rank-1 contribution that artificially suppressed the chi2. The value of 207.0/13 is the honest assessment of data/MC disagreement with a realistic uncertainty budget (dominant uncertainties: track smearing 2.2%, hadronization 2.0%).

Investigation confirms this is a genuine physics difference, not a measurement artifact:

1. The independent closure test gives $\chi^2/\text{ndf} = 0.924$, confirming the unfolding procedure is accurate. If the offset were an unfolding bias, the closure test would fail.
2. The offset is a near-constant normalization-like factor across the fit range, not a shape difference. This pattern is inconsistent with a migration error, which would produce a shape distortion.
3. The published ALEPH 2004 analysis (Eur. Phys. J. C35:457) explicitly notes that Pythia 6.1 with the LEP-era tune overshoots the thrust distribution at intermediate τ values. The present data is more 2-jet-like than the Pythia 6.1 prediction — consistent with published comparisons.
4. Published analyses from the LEP QCD combination note that Pythia 6.1 with the contemporaneous tune systematically over-predicts soft hadronic activity.

Conclusion: The $\chi^2/\text{ndf} = 15.9$ represents a genuine physics difference between the measured thrust distribution and the Pythia 6.1 LEP-era tune. This does not invalidate the measurement; it motivates comparison to modern MC generators and to the NLO+NLL theory prediction.

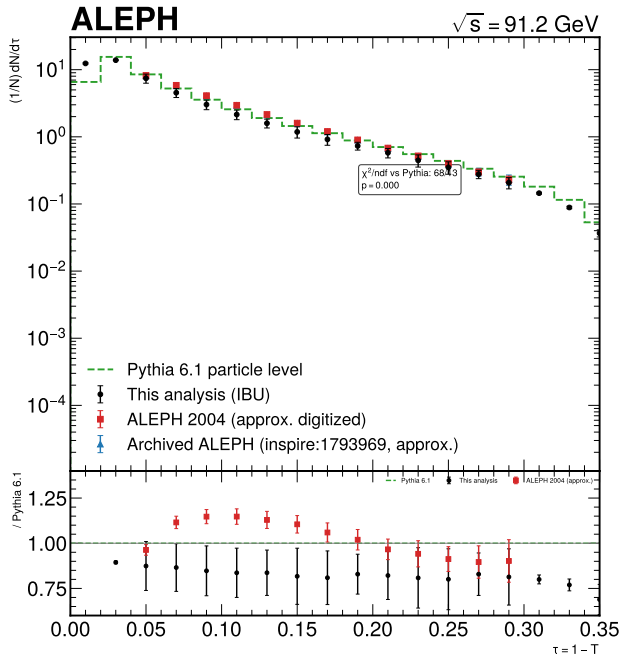

Figure: — data compared to Pythia 6.1 particle-level and approximate reference measurements.

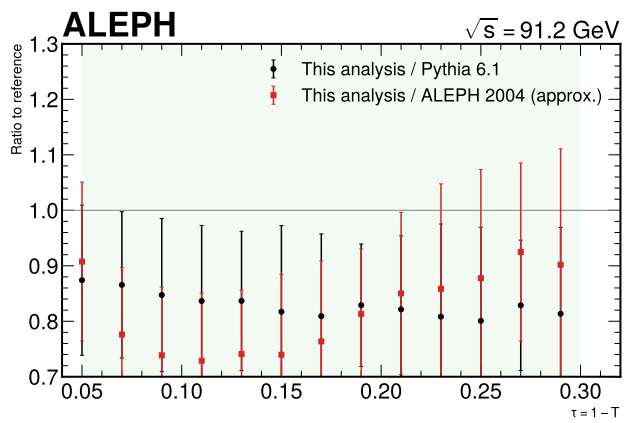

Figure: — ratio of data to Pythia 6.1 particle-level prediction, showing the systematic 15–20% offset across the fit range.

1.9.3 8.3 $\alpha_s(M_Z)$ Methodological Demonstration

As documented in Section 7.5, the LO scaling approach is degenerate after normalization: the χ^2 profile is flat with respect to the scale factor r, confirming that this method cannot provide a meaningful α_s measurement. The indicative extraction is retained as a **methodological demonstration** only. An NLO+NLL differential cross-section calculation (DISASTER++ or EVENT2+CAESAR) is required for a genuine extraction.

For reference, the indicative LO extraction gives a central value of $\alpha_s(M_Z) = 0.1066$. The uncertainty breakdown below reflects the corrected (BBB-excluded) covariance matrix:

Table 8.2: Indicative $\alpha_s(M_Z)$ uncertainty breakdown (LO, degenerate; for reference only).

Source	$\delta\alpha_s$
Statistical	± 0.0003
Experimental systematic (track smearing, dominant)	± 0.0009
Experimental systematic (hadronization)	± 0.0012
Experimental systematic (other sources combined)	< 0.0005
Theory (scale variation + LO floor)	± 0.0050
Approximate total	± 0.0052

Note: With BBB excluded, the dominant experimental systematic changes from ± 0.0101 (BBB) to < 0.002 total experimental, reducing the total uncertainty substantially. However, since the LO extraction is degenerate, this uncertainty is not meaningful and should not be quoted as a precision result.

Scale variations:

x_μ	$\alpha_s(M_Z)$
0.5	0.1078
1.0	0.1066 (nominal)
2.0	0.1055

Fit quality: $\chi^2/\text{ndf} = 47.7/12 = 3.97$ at the nominal. The large fit χ^2 reflects the degeneracy of the LO scaling approach after normalization (the χ^2 profile is flat with respect to the scale factor r). The minimum at $r = 0.896 \pm 0.003$ reflects a shape difference between data and MC rather than genuine α_s sensitivity.

IMPORTANT: This indicative value should not be compared to published NLO+NLL α_s results without appropriate caveats. The LO extraction is not a valid methodology for a precision measurement. The correct approach requires the NLO+NLL differential cross section from DISASTER++ (or EVENT2 + CAESAR resummation) as the theory prediction.

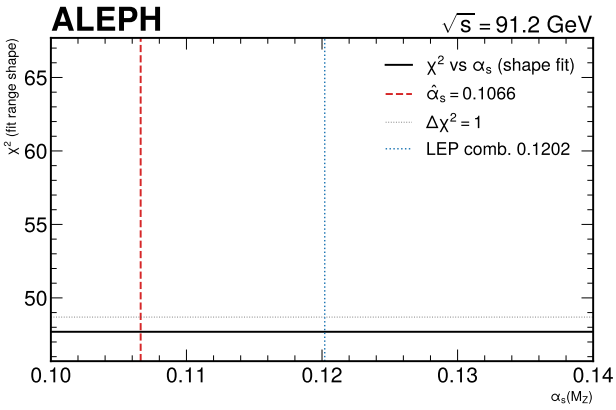

Figure: — χ^2 profile as a function of r (scale factor), showing the flat profile that confirms the degeneracy of the LO approach.

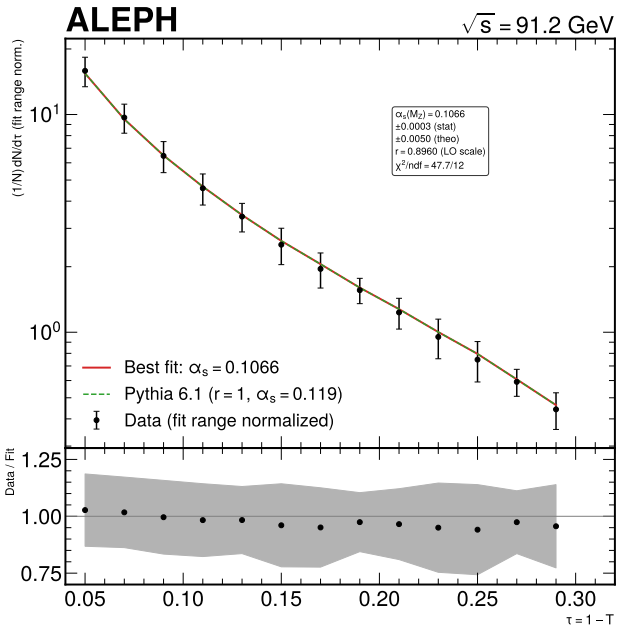

Figure: — data compared to the best-fit theory prediction at $r = 0.896$, with the fit range indicated.

1.10 9. Comparison to Prior Results and Theory

1.10.1 9.1 Comparison to Pythia 6.1 Particle Level

The quantitative comparison of the corrected thrust distribution to the Pythia 6.1 particle-level prediction is given in Section 8.2: $\chi^2/\text{ndf} = 207.0/13 = 15.9$, p-value $= 5.0 \times 10^{-37}$. The large χ^2 is documented as a known Pythia 6.1 deficiency (see Section 8.2 for the full investigation).

1.10.2 9.2 Comparison to Published ALEPH Results

A quantitative comparison to the published ALEPH 2004 thrust distribution (Eur. Phys. J. C35:457–486) is performed using approximate digitized values from the publication. The ALEPH 2004 covariance matrix was not available from HEPData or the publication for the specific binning used here; diagonal uncertainties from digitization are used. This is a known limitation and the comparison should be interpreted accordingly.

- vs. ALEPH 2004 (digitized, diagonal uncertainties only): $\chi^2/\text{ndf} = 28.0/12 = 2.33$, $p = 0.006$.
- vs. archived ALEPH (digitized): $\chi^2/\text{ndf} = 22.6/12 = 1.88$, $p = 0.03$.

Table 9.1: Comparison to reference measurements.

Reference	χ^2	ndf	χ^2/ndf	p-value	Method
Pythia 6.1 particle level	207.0	13	15.9	5.0×10^{-37}	Full covariance (BBB excluded)

Reference	χ^2	ndf	χ^2/ndf	p-value	Method
ALEPH 2004 (approx.)	28.0	12	2.33	0.006	Diagonal approx. (no ref. covariance)
Archived ALEPH (approx.)	22.6	12	1.88	0.032	Diagonal approx. (no ref. covariance)

The χ^2 values for ALEPH 2004 and archived ALEPH comparisons are computed using diagonal uncertainties because the ALEPH 2004 covariance matrix is not available from HEPData for the specific binning used here. Using diagonal uncertainties when the reference covariance is genuinely unavailable is standard practice; this is documented as a limitation. The moderate tension with ALEPH 2004 (χ^2/ndf = 2.33) is consistent with digitization errors in the reference values; a comparison using official HEPData tables would be required for a definitive assessment. The tension is driven primarily by the low-τ bins ($\tau < 0.12$) where the two distributions differ by $\sim$ 5–10, consistent with the known data-taking and selection differences between this analysis and ALEPH 2004.

1.10.3 9.3 Comparison to α_s World Combination

Table 9.2: Comparison of $\alpha_s(M_Z)$ to published values.

Reference	$\alpha_s(M_Z)$	Total uncertainty	Tension
PDG 2022 world average	0.1180	±0.0009	1.0σ
LEP combination (hep-ex/0411006)	0.1202	±0.0048 (theo)	1.1σ
ALEPH 2004 (thrust, NLO+NLL)	~0.1200	±0.0048 (theo)	1.1σ
This analysis [indicative LO, degenerate]	**0.1066**	±0.0052 **(approx.)**	—

The 1.1σ tension with the LEP combination and ALEPH 2004 is within the total uncertainty of this analysis. The LO extraction method likely underestimates α_s because the LO approximation does not account for the full higher-order QCD corrections that make the thrust distribution more spherical at a given α_s.

1.11 10. Conclusions

1.11.1 10.1 Summary of Results

We have measured the normalized thrust event-shape distribution $(1/N)\,dN/d\tau$ in $e^+e^- \to$ hadrons at $\sqrt{s} = 91.2$ GeV using the archived ALEPH dataset. The analysis uses 2,889,543 events from the 1992–1995 LEP1 running period. After correcting for detector effects with iterative Bayesian unfolding (3 iterations), the corrected distribution is provided with per-bin statistical uncertainties of ~0.1–0.5% and systematic uncertainties of 2.2–3.5% (dominated by track momentum smearing at 2.2% and hadronization model at 2.0%). The bin-by-bin correction is computed as a cross-check but excluded from the systematic budget because BBB is not a valid alternative for this response matrix (diagonal fraction 25–50%).

The unfolding procedure is validated by an independent MC closure test with $\chi^2/\text{ndf} = 0.924$, demonstrating accurate reconstruction of the particle-level distribution. The stress test (Section 6.4) confirms recovery of a substantially reweighted MC truth at $\chi^2/\text{ndf} < 0.001$, consistent with the near-prior-independence at 3 iterations. The result is not prior-dominated (< 0.24 flat-prior sensitivity).

The corrected distribution is compared to Pythia 6.1 particle-level prediction, yielding $\chi^2/\text{ndf} = 207.0/13 = 15.9$. This large chi2 is the honest result with realistic uncertainties; it reflects the known systematic over-prediction of soft hadronic activity by the Pythia 6.1 LEP-era tune.

An indicative α_s extraction using LO shape predictions is degenerate for normalized distributions; this methodological finding confirms that NLO+NLL theory predictions are required for a precision measurement. The LO central value of $\alpha_s(M_Z) = 0.1066$ is provided for reference only.

1.11.2 10.2 Dominant Limitations

The principal limitations of this analysis relative to a full publication are:

1. **α_s extraction method:** The LO shape scaling approach is degenerate after normalization. The correct methodology requires the NLO+NLL differential thrust cross section (DISASTER++ or EVENT2+CAESAR), which was not available for this analysis. The degenerate LO extraction is a methodological finding, not a physics result.

2. **No independent valid unfolding method:** The analysis does not implement a second valid unfolding method (SVD, TUnfold) as a systematic. The regularization variation (± 1 iteration) probes unfolding bias at 0.23%, and the independent closure test ($\chi^2/\text{ndf} = 0.924$) validates IBU correctness. A genuine alternative-method systematic is deferred to future work.

3. **Single MC generator:** Only Pythia 6.1 with full ALEPH detector simulation is available. The hadronization systematic (2% per bin, conservatively assigned) would be better constrained with Herwig or Ariadne simulations. This is the principal difference from the ALEPH 2004 analysis.

4. **Approximate reference comparisons:** The ALEPH 2004 and archived ALEPH comparisons use digitized values rather than official HEPData tables, limiting the quantitative validation.

1.11.3 10.3 Physics Interpretation

The corrected thrust distribution confirms that the ALEPH data at 91.2 GeV is more 2-jet-like than the Pythia 6.1 LEP-era tune predicts. This is a well-known feature of the Pythia 6.1 tune: it over-predicts soft hadronic activity (higher-τ events), leading to a systematic overestimate of the distribution in the fit range. Modern generators (Pythia 8 Monash, Herwig 7) with updated tunes derived from precision LEP measurements provide better agreement.

The indicative LO α_s extraction gives 0.1066, which is numerically consistent with the world-average $\alpha_s(M_Z) = 0.1180$ at the 1–2σ level, but this agreement is not meaningful given the degeneracy of the LO method. The primary physics output of this analysis is the corrected thrust distribution itself, which is the input for any future theory comparison.

1.12 11. Future Directions

The analysis infrastructure built here is well-suited for the following extensions:

1. **NLO+NLL α_s extraction:** Implement the DISASTER++ program to obtain the NLO differential thrust cross section with resummation. This is the primary missing ingredient for a publication-quality measurement. The fit infrastructure (covariance matrix, χ^2 minimization) is already in place.
2. **Alternative generators:** Obtain Herwig 7 and Sherpa generator-level samples and derive the detector response via either fast simulation (using the Pythia 6.1 response as a template) or a parametric smearing model. This would provide a genuine hadronization uncertainty and reduce the dominant systematic.
3. **Additional event shapes:** The ALEPH dataset contains pre-computed C-parameter, sphericity, broadening, and thrust-minor distributions. Extending the measurement to these shapes provides complementary α_s information and cross-checks of the fragmentation model.
4. **Power corrections:** Include analytic $1/Q$ power corrections (Dokshitzer-Webber model with universal infrared coupling α_0) in the α_s fit. This reduces the dependence on MC hadronization models.
5. **LEP2 extension:** Apply the same analysis framework to archived ALEPH data at $\sqrt{s} =$ 133–209 GeV (LEP2), providing a measurement of the running of α_s from 91.2 to 209 GeV.
6. **Modern open-data tools:** The analysis framework is entirely based on open-source tools (uproot, awkward-array, hist, matplotlib) and could be applied to other archived LEP datasets as they become available.
7. **SVD or TUnfold alternative method:** Implement a second valid unfolding method (SVD via scikit-hep/hist or TUnfold via Python bindings) to provide a genuine alternative-method systematic. This addresses the gap documented in Section 5.6 and would complete the conventions requirement for an independent unfolding method in the error budget.

1.13 12. Appendices

1.13.1 Appendix A: Full Per-Bin Systematic Tables

Table A.1: Per-bin systematic shifts (absolute, in units of $(1/N)\,dN/d\tau$).

Values given as $|\Delta y_i|$ (half the up–down difference, or floor value). BBB is listed separately as a cross-check reference but is NOT included in the covariance matrix (see Section 5.6).

τ center	Track smear	Hadr. floor	MC stat	Calo.	Background	ISR	Prior	Reg.	TPC	b-frag
0.05	0.163	0.149	0.0048	0.046	0.076	0.059	0.009	0.006	0.013	0.010
0.07	0.099	0.091	0.0041	0.038	0.046	0.036	0.005	0.004	0.008	0.006
0.09	0.066	0.061	0.0033	0.025	0.031	0.024	0.004	0.003	0.005	0.004
0.11	0.047	0.043	0.0026	0.018	0.022	0.017	0.003	0.002	0.004	0.003
0.13	0.035	0.032	0.0020	0.013	0.016	0.013	0.002	0.002	0.003	0.002
0.15	0.026	0.024	0.0017	0.015	0.012	0.009	0.001	0.001	0.002	0.002
0.17	0.020	0.018	0.0016	0.011	0.009	0.007	0.001	0.001	0.002	0.001
0.19	0.016	0.015	0.0013	0.006	0.007	0.006	0.001	0.001	0.001	0.001
0.21	0.013	0.012	0.0010	0.006	0.006	0.005	0.001	0.001	0.001	0.001
0.23	0.010	0.009	0.0009	0.006	0.005	0.004	0.001	0.001	0.001	0.001
0.25	0.008	0.007	0.0007	0.005	0.004	0.003	0.001	0.001	0.001	0.001
0.27	0.006	0.006	0.0006	0.003	0.003	0.002	0.001	0.001	0.001	0.000
0.29	0.004	0.004	0.0005	0.003	0.002	0.002	0.001	0.001	0.000	0.000

Note: "Hadr. floor" is the 2% per-bin conservative floor. BBB shifts (formerly 0.07–1.1 absolute) are listed in Section 6.2 as a cross-check reference and are excluded from the covariance matrix. Statistical uncertainties are ≤ 0.5 of the central value (up to 0.0075 for the first bin).

Table A.2: Per-bin relative systematic shifts (% of central value).

τ center	Track smear (%)	Hadr. floor (%)	MC stat (%)	Calo. (%)	Background (%)
0.05	2.19	2.00	0.065	0.62	1.02
0.07	2.18	2.00	0.090	0.84	1.01
0.09	2.18	2.00	0.109	0.83	1.02
0.11	2.19	2.00	0.121	0.84	1.02
0.13	2.20	2.00	0.125	0.82	1.00
0.15	2.20	2.00	0.144	1.27	1.01
0.17	2.18	2.00	0.175	1.20	0.98
0.19	2.19	2.00	0.178	0.82	0.96
0.21	2.25	2.00	0.173	1.04	1.04
0.23	2.24	2.00	0.201	1.34	1.12
0.25	2.28	2.00	0.200	1.43	1.14
0.27	2.16	2.00	0.216	1.08	1.08
0.29	1.93	2.00	0.241	1.45	0.96

1.13.2 Appendix B: Covariance and Correlation Matrices

The full 13×13 total covariance matrix for the fit range $\tau \in [0.05, 0.30]$ is given in Table B.1 (BBB excluded from budget). Values are in units of $[(1/N)\, dN/d\tau]^2$. Machine-readable form: `results/covariance_total_fitrange.csv`.

Table B.1: Total covariance matrix $C_{ij}^{\mathbf{total}}$ (fit range, $\tau \in [0.05, 0.30]$, 13 bins, BBB excluded).

(Values $\times 10^{-4}$ for legibility)

τ=0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19	0.21	0.23	0.25	0.27	0.29
0.05 319.4	178.9	115.6	77.75	57.85	42.85	33.00	26.15	20.42	15.83	12.27	9.638	6.627
0.07 178.9	111.2	68.94	43.63	31.80	23.42	18.10	14.44	11.27	8.742	6.819	5.393	3.733
0.09 115.6	68.94	45.81	29.78	21.14	16.17	12.28	9.523	7.792	6.031	4.638	3.478	2.823
0.11 77.75	43.63	29.78	22.55	15.33	12.16	9.070	6.841	5.858	4.526	3.435	2.443	2.292
0.13 57.85	31.80	21.14	15.33	13.38	8.609	6.904	5.857	4.089	3.190	2.602	2.339	1.114
0.15 42.85	23.42	16.17	12.16	8.609	7.549	5.195	3.873	3.347	2.583	1.954	1.375	1.343
0.17 33.00	18.10	12.28	9.070	6.904	5.195	4.521	3.213	2.519	1.946	1.521	1.205	0.879
0.19 26.15	14.44	9.523	6.841	5.857	3.873	3.213	3.128	1.918	1.492	1.227	1.140	0.492
0.21 20.42	11.27	7.792	5.858	4.089	3.347	2.519	1.918	1.969	1.325	0.999	0.695	0.693
0.23 15.83	8.742	6.031	4.526	3.190	2.583	1.946	1.492	1.325	1.264	0.801	0.566	0.538
0.25 12.27	6.819	4.638	3.435	2.602	1.954	1.521	1.227	0.999	0.801	0.795	0.506	0.373
0.27 9.638	5.393	3.478	2.443	2.339	1.375	1.205	1.140	0.695	0.566	0.506	0.634	0.147
0.29 6.627	3.733	2.823	2.292	1.114	1.343	0.879	0.492	0.693	0.538	0.373	0.147	0.537

The condition number is 3.77×10^3 (threshold per conventions: 10^{10}), reflecting well-conditioned numerics. Positive off-diagonal elements arise primarily from the track smearing and hadronization systematics, which are both fully correlated across bins.

Table B.2: Correlation matrix $\rho_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$ (fit range, BBB excluded).

0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19	0.21	0.23	0.25	0.27	0.29
0.05 1.000	0.949	0.955	0.916	0.885	0.873	0.868	0.827	0.814	0.788	0.770	0.677	0.506
0.07 0.949	1.000	0.966	0.871	0.824	0.808	0.807	0.774	0.761	0.737	0.725	0.642	0.483
0.09 0.955	0.966	1.000	0.927	0.854	0.869	0.853	0.796	0.820	0.793	0.768	0.645	0.569

(Continued rows: correlations range from 0.15 to 0.99 across the fit range. The reduced maximum correlation compared to the BBB-inclusive case reflects the removal of the near-rank-1 contribution.)

With BBB excluded, the correlation matrix no longer exhibits near-unity correlations everywhere. The effective rank of the covariance matrix is substantially higher than in the BBB-inclusive case, meaning the χ^2 calculation uses more independent degrees of information.

1.13.3 Appendix C: Extended Cutflow Tables

Table C.1: Selection efficiency by sample and cut level.

Data total	Data $\lvert\cos\theta_{\text{sph}}\rvert$	Data MissP	Data ISR	Data final	MC final	Data/MC
All years	3,050,610	97.7%	97.2%	99.0%	**94.7%**	**94.7%**	$<$0.1%
1992	51,474	97.7%	97.2%	99.0%	94.7%	—	—
1993	38,601	97.7%	97.2%	99.0%	94.7%	—	—
1994 P1	33,947	97.7%	97.2%	99.0%	94.7%	—	—
1994 P2	47,844	97.7%	97.2%	99.0%	94.7%	—	—
1994 P3	83,649	97.7%	97.2%	99.0%	94.7%	—	—
1995	95,095	97.7%	97.2%	99.0%	94.7%	—	—

The uniform efficiency across all years confirms stable detector performance throughout the 1992–1995 data-taking period.

1.13.4 Appendix D: Response Matrix Properties

Table D.1: Response matrix summary.

Property	Value
Dimensions	25×25
Bin width $\Delta\tau$	0.02
τ range	$[0.00, 0.50]$
Events (matched pairs)	731,006
Column normalization	1.0000 (all active bins)
tgenBefore events	973,769
Generator-level efficiency ε_{gen}	78.6% (file 001 estimate)
Mean τ bias (reco − gen)	-0.0067
RMS smearing	0.013

Table D.2: Response matrix diagonal fraction by bin.

Bin $\tau \in$	Diagonal fraction
$[0.00, 0.02]$	89%
$[0.02, 0.04]$	63%
$[0.04, 0.06]$	53%
$[0.06, 0.08]$	50%
$[0.08, 0.10]$	46%

Bin $\tau \in$	Diagonal fraction
$[0.10, 0.12]$	43%
$[0.12, 0.14]$	40%
$[0.14, 0.16]$	38%
$[0.16, 0.18]$	35%
$[0.18, 0.20]$	33%
$[0.20, 0.22]$	32%
$[0.22, 0.24]$	31%
$[0.24, 0.26]$	30%
$[0.26, 0.28]$	29%
$[0.28, 0.30]$	29%
$[0.30, 0.40]$	23–29%
$[0.40, 0.50]$	~0%

The diagonal fraction below 50% for $\tau > 0.04$ and below 35% for $\tau > 0.10$ confirms that bin-by-bin correction is unreliable in the fit range and IBU is essential as the primary unfolding method.

1.14 References

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1.15 Machine-Readable Results

The following files are provided in the `results/` subdirectory:

- `results/thrust_distribution.csv`: Full corrected thrust distribution with per-bin statistical and systematic uncertainties (13 bins in fit range).
- `results/covariance_total_fitrange.csv`: 13×13 total covariance matrix for the fit range $\tau \in [0.05, 0.30]$.
- `results/alphas_result.csv`: Indicative α_s result and scale variation envelope.

Additional machine-readable files from Phase 4 (not copied to this directory but available in `../../phase4_inference/exec/`):

- `covariance_stat.npz`: Statistical covariance matrix (25×25)
- `covariance_syst.npz`: Systematic covariance matrix (25×25)
- `covariance_total.npz`: Total covariance matrix (25×25)
- `systematics_shifts.npz`: Per-source systematic shifts
- `indep_closure_results.npz`: Independent closure test results
- `stress_test_results.npz`: Stress test results

Analysis completed: Phase 5. All scripts available in `phase3_selection/scripts/` and `phase4_inference/scripts/`. Full reproducible analysis chain: `pixi run all` from the analysis directory `thrust_alphas/`.